

University of Padova

Department of Mathematics

Master Thesis in Data Science

An efficient latent representation of music based on graph neural networks

Supervisor

Antonio Rodà

University of Padova

Co-supervisor

Matteo Spanio

University of Padova

Master Candidate

Giovanni Chessa

Academic Year

2025-2026

Abstract

Every neural audio codec in use today encodes music at a constant rate: the time–frequency plane is cut into units whose shape is fixed by the architecture before any audio is heard, and none of these systems can change the shape of the units it encodes. This thesis asks whether a representation of music can be built the other way round, by letting the signal itself decide how its own plane is divided. The substrate that makes the question expressible is a graph. Contracting neighbouring regions into a single unit — coarsening — is a structural operation that a graph supports natively and a grid cannot express, and it is the reason graph neural networks are used here. Until now they had reached audio only for discriminative tasks, where nothing has to be reconstructed; whether a graph representation of music can be inverted back into sound was a question that, to our knowledge, had not been posed.

The thesis answers it in the affirmative and measures what the answer costs. A spectrogram read as a graph and coarsened by its own signal reconstructs better, at a matched number of tokens, than a content-blind grid sharing every other component, and reaches a quality the grid does not match with seven times as many tokens — with a model of 654 850 parameters, a factor of twenty-seven below the closest published system. Fifty-two listeners in a blind test prefer it to the rate-matched control, and their ratings validate the perceptual metric the comparisons rest on. What decides the outcome is not adaptivity as such but the merge criterion: coarsening paid when the cost assigned to a merge approximated the error that merge causes, and lost when it did not. Inside the trained model, the tokens that cover silence collapse to a single connected region per clip, with two and a half times less variance than those of the note: the model arrives at a codeword for silence without being asked for one, and that is where the rate is saved. The efficiency claimed is a property of the representation and is counted in tokens, not in transmitted bits; the cost of the partition is measured and reported beside it. What this work places is a first stone: the first reconstruction model built on the coarsening of a graph.

Contents

Abstract iii

List of figures vii

List of tables ix

1 Introduction 1
- 1.1 Introducing the work . . . 2
- 1.2 Objectives and Contributions . . . 5

2 Related Work 11
- 2.1 Music AI . . . 11
- 2.2 Codec models . . . 13

3 Methodology 19
- 3.1 Preprocessing . . . 19
- 3.2 The encoder and the adaptive coarsening . . . 22
- 3.3 Decoder . . . 27
- 3.4 Loss . . . 28
- 3.5 Parameters discussion . . . 29

4 Experiments 31
- 4.1 Ablations . . . 32
- 4.2 Measuring quality, and counting the partition . . . 36
- 4.3 The subjective listening test . . . 41
- 4.4 Probing the learned representation . . . 43

5 Results 47
- 5.1 Results on the model . . . 48
  - 5.1.1 Reconstruction on the full corpus . . . 48
  - 5.1.2 Distributional evaluation on the test split . . . 51
  - 5.1.3 The listening test . . . 54
- 5.2 The latent space . . . 56
  - 5.2.1 What the bottleneck learned to contain . . . 56
  - 5.2.2 What this analysis is not . . . 64

6 Discussion **65**
6.1 What the work establishes 65
6.2 Why the graph was the right substrate 66
6.3 Two currencies, both honest 67
6.4 The first stone 68

7 Conclusion **71**

References **73**

Listing of figures

1.1 How much of a note is silence 3

2.1 Three ways of partitioning the time–frequency plane 16

3.1 The GAAT pipeline 20
3.2 The time–frequency graph 22
3.3 The encoder block and the reach of a Chebyshev filter 24
3.4 Total variation against Ward's linkage 26
3.5 The three stages of the decoder 28

4.1 The two terms of the Fréchet distance 37
4.2 Why a paired measurement is needed alongside the Fréchet distance 38

5.1 Partitions produced by the two assignments 49
5.2 Rate–distortion curve, with the adaptive operating point 50
5.3 Information carried by each latent dimension 57
5.4 The dominant latent dimension as a gain 58
5.5 The pitch control on neighbourhood coherence 60
5.6 The latent space with and without the level profile 62

Listing of tables

3.1 Parameter count by component, with the variational readout enabled. 29

4.1 Merge criterion, at a rate pinned by the node ceiling. The cost function is varied against the feature space in which distances are measured. 33

4.2 Harmonic edges against the time–frequency grid alone, at two node ceilings. At 8192 the grid-only arm wins having trained for less than half as long. . . . 34

4.3 Collapsing the silent region into K super-nodes before coarsening. No configuration reaches the model that gives silence no special treatment, and all of them spend more tokens. 35

4.4 Cost of transmitting the partition, in kilobits per second, over 128 clips of the test split. The estimators for the uniform arms are italicised because they are hypothetical: a fixed grid is defined by two integers known to both ends, so its partition costs nothing and its total is the payload alone. 39

5.1 Node-to-token assignment on the full NSynth validation set, at a comparable token count. The three models share encoder, decoder, objective and $\beta = 10^{-3}$; only the way graph nodes become tokens differs. Adaptive coarsening under the total-variation criterion loses to fixed-grid pooling. Adding the cluster-size term to the merge cost reverses the verdict. 48

5.2 Distortion against information rate. The divergence of the variational bottleneck is a rate in nats that requires no quantisation. Uniform pooling spends 43 401 nats per clip without matching the quality that the adaptive method reaches with 5024. 51

5.3 Fréchet distance and paired cosine over CLAP embeddings, on the 4096 clips of the test split. The Griffin–Lim row receives the exact magnitude of the original and errs only in phase; it is the floor imposed by phase reconstruction, not a competing system. 51

5.4 Ward against every other system. Differences in Fréchet distance, negative meaning Ward is closer to the originals, with 95% intervals over 200 bootstrap resamples evaluated on the same clips. 52

5.5 EnCodec at increasing rate, on the same clips, with the adaptive model placed on the same scale. 53

5.6 Mean opinion score on the five-point scale, averaged over the seven clips within each participant and then over the fifty-two participants who passed the screening, with the difference from the adaptive model computed *within* each participant. Intervals are percentile bootstrap intervals over 10 000 resamples of the participants; the last column counts the participants preferring the adaptive model, ties excluded. 54
5.7 Correlation between the objective measurements and the mean opinion score over the thirty-five clip-by-condition cells. The right-hand block removes the seven cells of the hidden reference and is the one that matters. 55
5.8 Timbral attributes, area under the curve on an instrument-disjoint split. The last two columns are the measurement that matters: removing pitch costs almost nothing, removing the structured level profile costs a great deal. The picture is mixed by attribute and the mean alone would misrepresent it. The first two columns come from the pitch control and the last three from the level control, run separately; the intact representation scores 0.792 in the first and 0.787 in the second, which is the run-to-run variation of the coarsener rather than a difference of protocol. 61

1 Introduction

The system worth building is not the one that compresses most. It is the one small enough to run where music is actually made — on a laptop, inside an instrument, under a performer's hands. That constraint is not a limitation accepted late in the design; it is the premise this work starts from, and it changes what counts as a flaw. Every token the representation emits is computation the decoder must spend, and every parameter is memory the machine must hold. Against a budget of that kind a token spent on nothing is not an inefficiency: it is a token taken from somewhere else.

A recording of a musical note is not informationally uniform. The attack of a plucked string, the steady portion of the tone that follows it, and the decay into near-silence differ by orders of magnitude in how much must be said about them to describe them faithfully. Neural audio codecs, however, say the same amount about all three. Architectures such as SoundStream [1], EnCodec [2] and DAC [3] encode a waveform by striding a convolutional network over it, emitting one latent vector every fixed number of samples. The resulting token stream has a constant rate: a quantisation of *duration* rather than of *content*, and a partition of the time–frequency plane whose shape is fixed by the architecture before any audio is seen.

This thesis asks whether a graph can do better — whether music can be given an efficient latent representation by letting the signal decide how its own plane is divided. The question is one of representation rather than of transmission, and the answer this work gives is a mechanism: a spectrogram read as a graph and *coarsened*, merging adjacent regions under a criterion until what remains is a shorter sequence of tokens. Coarsening is the reason the graph is here.

It is a structural operation that a graph affords and a grid does not, because a grid has no way to contract unevenly, and it is the operation on which everything in the following chapters depends. Graph neural networks have been applied to audio almost only for classification, where the phase is discarded and nothing has to be reconstructed; using them as the encoder of a reconstruction model, and using the coarsening of the graph as the mechanism that forms the representation, has not to our knowledge been attempted before.

1.1 Introducing the work

The starting hypothesis

Silence is where that waste is easiest to see, and it is the form in which the problem was first posed here. A fixed-rate encoder cannot avoid spending representational capacity on regions that carry almost no information, and silence is the extreme case rather than a marginal one: over the full validation set of the corpus used here, 62.2% of the time–frequency bins fall below the silence threshold. On the balanced subset on which most of the ablations were run the percentage is 47.8%, and a fixed-grid pooling operator assigns 47.7% of its tokens to those bins. The near-identity of the two numbers is explanatory: the encoder treats signal and silence in the same way if not trained to do otherwise.

The imbalance of signal and silence also depends strongly on the instrument, as Figure 1.1 shows. Measured over forty clips per family, the silent fraction averages 81% for plucked guitar, whose energy has collapsed well before the recording ends, and 8% for sustained voice, which fills the window. It varies almost as much *within* a family as between families – from 56 to 99% across guitar clips alone – because it depends on the pitch and the velocity of the note as much as on the instrument. A constant token rate is therefore not uniformly wasteful: it is nearly harmless on sustained material and severely wasteful on percussive or decaying material, and no single fixed rate is right for both. An encoder able to allocate its budget according to the signal would, on this reasoning, recover a large fraction of its tokens at no cost in quality.

That was the hypothesis, and it did not survive contact with the measurements in the form in which it was posed. The models that collapse silence most aggressively are the ones that reconstruct worst, and the configuration that ultimately performs best spends, per unit area, more tokens on silence than on signal. The mechanism behind the failure, however, is not that the silent regions were carrying information after all; but rather that forcing active regions to be split into components that can no longer be merged with one another – several hundred

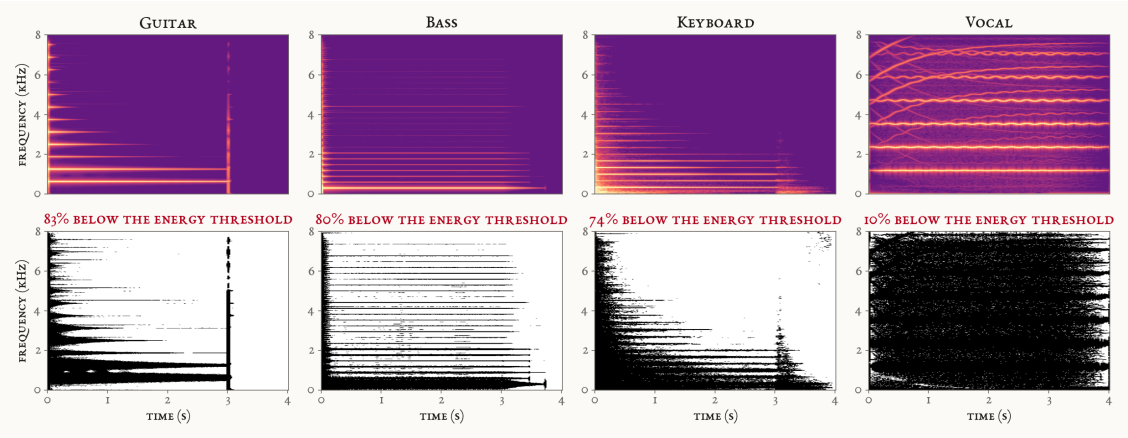


Figure 1.1: How much of a note is silence, and how differently it is distributed. The top row shows the log-magnitude spectrogram of one clip per family, chosen at the median silent fraction of forty clips sampled from that family; the bottom row marks in black the bins above the silence threshold used throughout this work. A plucked note occupies a few harmonic ridges and stops halfway through the window, leaving most of the plane empty; a sustained voice fills it. A fixed-rate encoder assigns the same number of tokens to all four.

connected components per clip, where an unconstrained coarsener sees one – so the partition degrades in *shape* before it degrades in size, and the token count settles above the imposed ceiling rather than below it. At high rates, where budget is abundant, collapsing silence does pay, but the model becomes useless without a strong compression; therefore the idea of dividing the signal into active and silent regions might be useful, but in a different regime.

The hypothesis was therefore not so much refuted as found to be the wrong question. What proves to matter is not *where* the tokens are placed but *by which criterion* the aggregation is carried out, and the argument of this thesis moves accordingly: from silence, to the number of tokens per clip and to the merge criterion that determines it.

Why a graph

A convolutional encoder inherits the geometry of its input. Its receptive fields are rectangles, its strides are fixed in advance, and the partition of the time–frequency plane it induces is a regular grid whose shape is a property of the architecture rather than of the signal. Any adaptivity must then be added on top of a structure that does not itself admit it.

A graph carries no such geometry. It has no intrinsic dimension, no notion of a rectangle, and no fixed neighbourhood: it is defined entirely by which elements are declared adjacent to which. It can therefore be contracted unevenly, folding where the signal is redundant and remaining fine where it is not, and the resulting partition is constrained by nothing except the

data and the criterion used to build it. This flexibility is the reason a graph was chosen as the substrate of the representation, and the sense in which the latent code produced here is adaptive by construction rather than by post-processing.

The justification is structural and does not depend on which edges are drawn. Contracting a region into a super-node is an operation a graph supports natively and a grid cannot express: a grid can only be subsampled on a schedule fixed in advance, whereas a graph can be merged wherever its own contents suggest, and the result is still a graph on which the same machinery runs. That single affordance — coarsening — is what the method is built on. Which edges to draw is a further degree of freedom the substrate makes available, and this thesis explores only part of it: the time–frequency lattice throughout, and a harmonic prior that did not survive its ablation. Chapter 6 returns to the edges that remain untried, because the space they open is a property of having chosen a graph in the first place.

Concretely, each bin of the short-time Fourier transform becomes a node carrying its log-magnitude, and adjacency in time and frequency becomes an edge. A clip of four seconds sampled at 16 kHz, transformed with a window of 1024 samples and a hop of 256, yields a graph of 251 frames by 513 frequency bins, that is 128 763 nodes. The encoder contextualises each node with its neighbours; a coarsening procedure then repeatedly merges adjacent nodes into super-nodes, each of which becomes one token. Because the merging stops on a criterion rather than after a prescribed number of steps, the length of the token sequence is a property of the clip.

The propagation over that graph was measured rather than assumed. When it is switched off — each node then encoded from its own log-magnitude alone, with depth, normalisation, coarsener, decoder and loss left exactly as they were — the reconstruction error rises by 65%, fourteen times the run-to-run variation of the pipeline. The loss is not attributable to the smaller network that results: an arm that restores the parameter count by adding depth lands in the same place. Message passing over the time–frequency graph is therefore doing substantial work, and Section 1.2 places that result among the contributions.

The data

All experiments in this thesis use NSynth [4], a corpus of individually recorded musical notes of four seconds each, sampled at 16 kHz in mono. The choice makes the phenomenon under study unusually legible: every clip contains a single note with a well-defined attack and decay, so the non-uniformity of information over time is present in an almost idealised form, and the

silent fraction is large and varies systematically across instrument families.

It also imposes a limit that should be stated at the outset rather than discovered by the reader. NSynth consists of isolated notes, not of music in the sense of polyphonic, mixed, structured recordings. It is a testbed on which the mechanism can be isolated, and the conclusions of this thesis are conclusions about that testbed. However, working with single notes forces the model to focus on the qualitative aspects of the audio: the essence of the sound is often difficult to appreciate, but it has its artistic purpose. Chapter 3 describes the dataset and the preprocessing in full, and Chapter 6 returns to what would be required to extend the claims beyond it.

1.2 Objectives and Contributions

Objective

The objective of this work is a latent representation of audio that is efficient in two distinct senses. The first concerns the representation: at a given reconstruction quality it should cost fewer tokens, and fewer nats, than one produced by an encoder that partitions the signal by a fixed rule. The second concerns the model that computes it. The complete system used in this thesis has 654 850 parameters, against the 17.6 million of RAVE [5], a factor of 27. RAVE synthesises 48 kHz audio twenty times faster than real time on a laptop CPU. This is a design commitment and not a concession: a representation that requires a large model to be computed is efficient only in one of its halves, and the compact models that can be trained and run on modest hardware are, in our view, an interesting way to compete against the large models that dominate the literature.

The model, in outline

Audio is transformed into a short-time Fourier transform and the magnitude spectrogram is read as a graph. Six Chebyshev graph-convolution layers [6] contextualise each node. A coarsening procedure then merges adjacent nodes in rounds, following a matching strategy adapted from heavy-edge matching in graph partitioning [7], until a stopping criterion is met; each surviving super-node is read out as a token by a weighted mean over the nodes it absorbed, optionally through a variational bottleneck whose Kullback–Leibler divergence provides a rate in nats [8]. A graph decoder scatters the tokens back onto the fine graph and predicts the log-magnitude, from which the waveform is resynthesised by the Griffin–Lim algorithm [9].

Chapter 3 develops each stage.

Contributions

1. **A merge criterion that accounts for cluster size, and the demonstration that adaptivity alone does not pay.** The quantity minimised when two clusters are merged determines the entire partition, and the criterion originally used, a reduction in graph total variation, is blind to how large the clusters being merged already are: absorbing a cluster of a thousand nodes into another costs the same as merging two isolated bins. Under that criterion the adaptive coarsener *loses* to the fixed grid. Replacing it with a Ward-style cost [10], which weights the distance between cluster means by the sizes of the clusters and thereby approximates the error that the subsequent pooling will commit, reverses the verdict: at a matched token count the adaptive method reconstructs 1.38 times better than the fixed grid while transmitting 17% fewer nats. The ordering of the three arms is the result, not any one of them: adaptivity by itself does not pay. Here it paid when the merge cost approximated the pooling error, and lost when it did not.

2. **Quality that a fixed grid does not reach with seven times as many tokens, measured against a control that makes the claim falsifiable.** At roughly 420 tokens per clip the adaptive configuration reconstructs better than uniform pooling at 2907. The comparison is measured rather than interpolated, and it rests on a content-blind pooling operator that tiles the same graph into fixed rectangular blocks while sharing the encoder, readout, decoder and loss, so that a single variable separates the two branches: the assignment of nodes to tokens. Without such a control the word *adaptive* is not falsifiable, and it is what revealed that the criterion used for most of the project was losing. The factor holds under the training objective on the validation set, where it is 6.93, and independently under a distributional metric on the held-out test split that is the objective of neither system, where it is 6.7. It is a factor in *token count*: the accounting in transmitted bits is different, and is the subject of the last paragraph of this section.

3. **A measurement of what the graph convolution contributes.** Twenty-five experiment groups had probed the edges, the merge criterion and the input representation, but never the propagation, so the quantity on which the premise of this work rests had no number attached to it. Reducing the Chebyshev filter to its order-zero term turns each layer into a per-node linear map with no neighbour aggregation, while leaving depth, normalisation, residuals, coarsener and decoder identical. Quality degrades by 65%, against a measured noise floor of 4.5%. A third arm restores the removed parameters by adding depth and reproduces the degraded result rather than the intact one, which separates propagation from capacity. The result comes with a limit worth stating: the graph does not disappear under this ablation, since the coarsener still merges along real edges, so what is measured

is the value of contextualising a node with its neighbours, not the value of the graph as such.

4. **A listening test, and the validation of the objective metric against human hearing.** Fifty-six listeners rated seven clips under five blind conditions with a hidden reference, all tracks loudness-aligned. Adaptive coarsening is preferred to the uniform grid at matched rate, to EnCodec at $1.5\,\mathrm{kbit\,s^{-1}}$ and to the total-variation criterion, and the mean opinion score, the Fréchet distance and the paired cosine order the four systems identically. The result that outlives this thesis, however, is the other one: across the thirty-five clip-by-condition cells the CLAP cosine predicts the human rating at a correlation of 0.787 among degraded systems, where the log-mel distance manages -0.491. The metric this work moved to is the one that survives the case that matters, and that is shown with human data rather than argued. The preference itself is small and uneven, which is reported with it: it is less than a fifth of a point, it is decisive on two stimuli out of seven and reversed on one, and it does not survive the strictest post-screening threshold.

5. **A map of what the substrate does not yet give for free.** Several components expected to help do not, and each is reported with the mechanism that explains it, because in an exploratory setting a mechanism is what makes a negative result usable. Structure imposed on the graph in advance fares worse than structure the coarsener finds on its own: sealing the silent region into fixed super-nodes fragments the active region and never reaches the model that treats silence like everything else, and harmonic edges specified by hand reconstruct worse than the plain lattice, reproducibly, because merging bins at different frequencies flattens the spectral envelope. A convolutional stem degrades the merge decision and a perceptually weighted merge bonus has no measurable effect. The rate itself is the clearest instance: instrumented carefully, the token count turns out to vary by only nine to twelve per cent across clips, so what this work demonstrates is an adaptive *partition* and not yet an adaptive *rate*. The width of the harmonic affinity is nonetheless the one quantity in the model that admits a physical reading, converging from 50 to approximately 28 cents, an estimate of how far the partials of this corpus depart from exact harmonicity. In an exploratory setting these are results, and instrumenting a claim carefully enough to weaken it is worth as much as confirming it.

6. **An analysis of what the bottleneck came to contain.** The trained model is frozen and its code examined with linear probes, which yields an account of *why* the preceding results happened. Eighty-seven of the 128 latent dimensions are unused and one carries 45% of the information; that dimension is a logarithmic gain, so that adding a constant to it multiplies the amplitude of the reconstruction by 3.45. Silent tokens occupy a region with two and a half times less variance than active ones and form a single connected region per clip. The code the model arrived at is a magnitude code organised around silence, which is precisely what buys it the rate. This last contribution is exploratory: it

was designed after the results it explains, it produced one claim that fell under its own control, and nothing established above depends on it.

Scope and limitations

Several limitations bound the claims made here, and are stated at the outset so that the results of Chapters 4 and 5 are read correctly.

The only external system run here is a single pre-trained codec, evaluated at one operating point that is its lowest rate, so the comparisons that carry weight are the internal and controlled ones; the published figures of other systems appear in Chapter 2 as context, not as a baseline. Only one adaptive operating point was trained on the full dataset, so the defensible statement is that this point dominates the fixed-grid curve over the range that was measured, not that an adaptive curve dominates a uniform one. Every figure obtained under the training objective is a validation figure; the test split is used once, for the distributional evaluation and the listening test, both of which are measured with quantities the models were not trained on. No run was seeded, and the run-to-run variation measured between two identically configured trainings is 4.5%, which is the threshold below which no difference is interpreted as an effect. The model reconstructs magnitude only and recovers phase by iterative resynthesis, which places a ceiling on the achievable quality and makes waveform-domain metrics uninformative by construction.

One further limitation cuts directly against the comparison this thesis rests on, and it is measured rather than merely acknowledged. A token sequence does not suffice to reconstruct: the decoder must also know which grid nodes were merged into which super-node, and it cannot recompute that map, because the map derives from encoder features the decoder never receives. The fixed-grid control needs no such side information, since its partition follows from two integers known to both ends and costs exactly nothing. Under a realistic coder the adaptive partition costs about ten times the payload of the tokens it describes, which brings the true rate of the model to roughly $21.5\,\mathrm{kbit\,s^{-1}}$ rather than the 1.9 that the token count alone implies.

The consequence is stated here rather than left to be discovered. Measured in transmitted bits, the advantage of adaptive coarsening is between 1.0 and 1.6, not the factor of seven quoted above; the two figures are denominated in different units and neither supersedes the other. This thesis concerns a representation, whose natural unit is the token, and no claim is made anywhere in it that this model compresses better than a neural codec.

Organisation

Chapter 2 reviews neural audio codecs, discrete audio representations and the graph-coarsening literature on which the method draws, and states the sense in which this work is positioned with respect to them. Chapter 3 describes the method: the construction of the time–frequency graph, the encoder, the coarsening and readout, the decoder and the loss. Chapter 4 reports the experimental campaign, including the ablations that isolate each component. Chapter 5 presents the quantitative results against the content-blind control, the judgement of human listeners, and an analysis of the learned representation. Chapter 6 weighs what the work establishes against what it attempted and did not reach. Chapter 7 states what is carried away from it.

2
Related Work

Two bodies of work bound this thesis. The first is the broad field of machine learning applied to music, which supplies the reason why compact audio representations matter at all: nearly every system built there consumes a representation it did not itself design. The second is the literature on neural audio codecs, which is where such representations are actually produced, and against which the method of this thesis has to be positioned. The first is reviewed briefly and for context; the second is examined in detail, and organised by the problem each family of work addresses rather than by chronology.

2.1 Music AI

Generative systems and the representations they assume

Music has become one of the principal application domains of large-scale generative modelling, to the point that the field now has its own surveys of foundation models [11]. The recent literature spans a wide range of conditioning signals: text-to-audio systems built on flow-matching objectives that trade sampling steps for latency [12], image-conditioned generation extended from natural photographs to artworks [13], and conditional systems that reinterpret an existing piece while preserving its melodic identity [14]. Retrieval and understanding have followed a parallel trajectory, with contrastive frameworks aligning sheet music, performance signals, audio and multilingual text in a single space [15].

What matters here is not the diversity of these systems but a property they share, and it is the one Chapter 1 takes as its starting point: none of them designs the representation it consumes. Each reads a sequence of tokens or latent vectors from a separately trained audio autoencoder, and inherits its decisions about what to keep and what to discard. Two consequences run upward through the pipeline. The length of the sequence sets the cost of attention, and the structure of the tokens sets how hard the sequence is to model — the second being the reason a codec cannot be evaluated on reconstruction alone. This is the sense in which the present work, which is about tokenisation and not about generation, is relevant to the field surveyed here.

What is inside a learned representation

A second line of work asks what these representations actually contain, rather than what they enable. Controllable generation is usually justified by appeal to disentangled embeddings, described as separating structure from timbre or local from global factors; a recent probing study across several music audio models finds that the intended and the actual semantics of those embeddings frequently disagree, and that current unsupervised strategies fall short of producing genuinely disentangled factors [16]. The methodology of that study, which uses linear probes over frozen embeddings together with controls for informativeness and invariance, is close to the one adopted in Chapter 4 to examine the latent space of the model proposed here, and its conclusion is a useful caution: a representation that supports a task is not thereby a representation organised around the factors one hoped for.

Graph neural networks in audio

Graph neural networks have entered audio research along a narrow path. ATGNN (Audio Tagging Graph Neural Network) treats the spectrogram as a graph rather than as an image or a sequence of patches, on the argument that the receptive field of convolutional networks remains limited while transformers impose a rigid patch grid that cannot follow irregularly shaped audio events; it reaches performance comparable to transformer baselines on audio tagging with substantially fewer parameters [17]. GraFPrint builds a k-nearest-neighbour graph over a time–frequency representation and applies max-relative graph convolutions to produce audio fingerprints robust to ambient distortion [18].

A separate and larger strand applies graph neural networks to *symbolic* music, where a score is naturally a graph of notes joined by succession, simultaneity and voice leading; there the graph is given by the notation rather than constructed from a signal, and the tasks are analytical.

The audio-domain works start from the same premise as this thesis, namely that the time–frequency plane is better served by a structure that does not impose a rectangular geometry on it. Both, however, are *discriminative*: they map audio to a label or to an identifier, and the phase of the signal, along with everything else needed to reconstruct it, is discarded at the first layer. The question of whether a graph representation can be inverted back into audio does not arise for them, and to our knowledge has not been systematically posed elsewhere. That gap is the space this work occupies.

2.2 Codec models

The dominant paradigm: fixed-rate residual quantisation

The modern neural audio codec was consolidated by SoundStream, which combined a fully convolutional encoder–decoder with residual vector quantisation and trained it with a mixture of reconstruction and adversarial objectives [1]. EnCodec refined the recipe with a multiscale spectrogram discriminator, a loss-balancing mechanism and an entropy coder over the quantised indices [2], and the improved RVQGAN of the Descript Audio Codec pushed fidelity further, notably by addressing the codebook collapse that limits how many quantisation stages can be used productively [3]. All three descend from the discrete bottleneck introduced by VQ-VAE, where an encoder output is snapped to the nearest entry of a learned codebook and gradients are passed through the non-differentiable assignment by a straight-through estimator [19].

These systems differ in their discriminators, their losses and their codebook mechanics. They do not differ in the property that concerns this thesis. In each of them the encoder is a strided convolutional stack, so the number of latent vectors produced for a signal is a fixed function of its duration: a token every N samples, whatever those samples contain. Residual quantisation makes the *depth* of the code adjustable, since dropping later stages lowers the bitrate, but it leaves the *length* of the sequence untouched. Rate is controlled along the axis of how finely each frame is described, never along the axis of how many frames deserve description.

The consequence is a representation whose granularity is a property of the architecture rather than of the signal. A steady tone and a transient attack receive the same budget; so does silence. Whether this is wasteful, and whether recovering the waste translates into quality, is the empirical question this thesis takes up.

Continuous bottlenecks, and rate without quantisation

A second family avoids discrete codes. RAVE is a variational autoencoder over raw waveforms, trained in two stages: a first stage that learns the representation under a multiscale spectral distance and a Kullback–Leibler penalty, and a second in which the encoder is frozen and only the decoder is refined adversarially [5]. Its compactness is obtained after training rather than chosen before it, by discarding the latent directions that carry negligible posterior energy. Its encoder is nevertheless a strided convolutional stack producing a latent at a constant rate, so on the axis that matters here it belongs with the codecs above; it is the closest published system to the present work in objective and the furthest in mechanism, and for that reason it is used throughout this thesis as the reference point for model size and for what a convolutional encoder achieves.

CoDiCodec shows that the choice between the two families need not be made at all: a single trained autoencoder yields both continuous embeddings at roughly $11\,\mathrm{Hz}$ and discrete tokens at $2.38\,\mathrm{kbit\,s^{-1}}$, using finite scalar quantisation together with a dropout scheme over the quantiser, and supports parallel as well as autoregressive decoding [20].

The continuous family also supplies the measurement device used in this thesis. When the bottleneck is variational, the Kullback–Leibler divergence to the prior is an information rate in nats, adjustable through the weight β [8]. It is a rate in the information-theoretic sense without any codebook having been trained, which is what allows a claim of efficiency to be made and checked on two independent axes in Chapter 5, one counting tokens and one counting nats.

Making the rate a variable

The observation that a constant rate is wasteful is not new, and since 2025 it has produced a rapidly growing line of work, almost all of it in speech. Temporally Flexible Coding tunes the average frame rate of a neural speech codec by allocating frames according to the temporal entropy of the signal, spending more of them on information-dense regions and fewer on silence [21]. VARSTok segments speech into variable-length units with a temporal-aware density-peak clustering algorithm and folds the duration of each unit into the token index itself, reporting better reconstruction naturalness with up to 23% fewer tokens than a fixed $40\,\mathrm{Hz}$ baseline [22]. FlexiCodec adaptively merges semantically similar frames and exposes an average frame rate that can be chosen at inference time between 3 and $12.5\,\mathrm{Hz}$ [23].

These systems establish the premise of this thesis rather than compete with it, and the differences are worth stating precisely because they define what is left to do. All of them are *temporal*:

the unit that varies is the frame, the axis along which allocation is decided is time, and the frequency axis remains whole and undivided. Their adaptivity is one-dimensional, and it is usually driven by a signal external to the codec proper, whether an entropy estimate, a semantic similarity between adjacent frames or the features of a pretrained recogniser. All of them, moreover, address speech, where the units that motivate variable segmentation are phonetic and where a pretrained recogniser is available to supply them; music has no equivalent segmentation and no equivalent supervision.

The representation studied in this thesis is variable along both axes at once, because the object being partitioned is not a sequence of frames but a two-dimensional plane, and the unit is not a time interval but a connected region of that plane that may be arbitrarily shaped. Whether this additional freedom is worth its cost is precisely what Chapter 5 measures, and the answer turns out to depend far more on the criterion governing the partition than on the freedom itself.

The structure of the time–frequency plane

The most recent work has begun to treat the internal structure of the representation, rather than its fidelity, as the object of design; this is where the concerns of this thesis and those of the codec literature converge most closely.

HARP starts from the observation that residual quantisation treats all frequencies alike, so the codebooks of successive stages become spectrally entangled and truncating them removes an unpredictable mixture of frequency content. Parallel band decomposition fixes the entanglement but fragments the latent space and destroys cross-frequency coherence. HARP instead partitions the quantisation stages into frequency-ordered groups, each refining its own band while the decoder retains access to all lower ones, so that overtones are reconstructed in the context of their fundamentals; the change is confined to the training objective and leaves inference identical to standard RVQ [24]. BandTok approaches the same plane from the other side, representing each frame by Mel-band tokens drawn from a single shared codebook in place of a residual hierarchy, on the argument that the resulting two-dimensional token grid is physically interpretable and easier for an autoregressive model to consume than a flattened residual sequence [25].

Both accept a premise this thesis does not. HARP partitions the *codebook* along the frequency axis and BandTok partitions the *plane* into a regular band-by-frame grid; in both cases the partition is fixed in advance and is the same for every signal. What varies with the content is the value assigned to each cell, never the geometry of the cells themselves. The step this thesis

takes is to let the partition itself be a function of the signal, which requires a substrate on which an irregular partition is expressible at all.

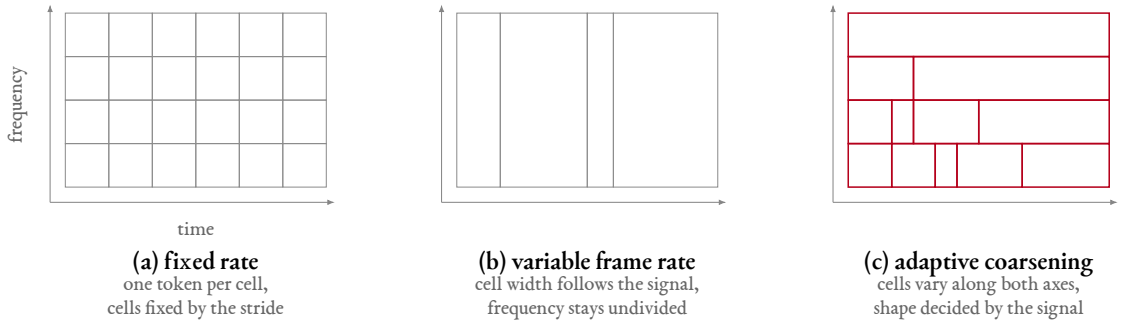


Figure 2.1: Three ways of partitioning the time–frequency plane, and the axis along which each of them adapts. Fixed-rate codecs impose a regular grid whose cells are a property of the stride. Variable-frame-rate codecs let the width of a cell follow the information density of the signal, but the cell still spans the whole frequency axis. The method of this thesis lets cells vary along both axes and take shapes that are not rectangles at all; the partition drawn here is schematic, and an actual one is shown in Chapter 5.

Graphs, graph signals and coarsening

That substrate is supplied by graph signal processing, which extends harmonic analysis to signals defined on irregular domains where the notions of translation and of uniform sampling that underpin classical processing do not apply [26, 27, 28]. A graph carries no intrinsic geometry: the neighbourhood of a node is whatever the edge set declares it to be, so a partition of the domain is not constrained to be a grid, and the notion of contracting the domain unevenly is native rather than bolted on.

Two ingredients of the method proposed here come from this literature. The first is spectral graph convolution: the Chebyshev polynomial filters of Defferrard et al. approximate spectral filtering on a graph without an explicit eigendecomposition, giving localised filters at a cost linear in the number of edges [6], and they are the operator used throughout this work. The second is graph coarsening. Multilevel graph partitioning reduces a graph by repeatedly matching and contracting adjacent vertices, and the heavy-edge heuristic of METIS, which prefers to contract the heaviest available edge at each step, is the strategy from which the coarsener of this thesis is adapted [7]. A more recent literature places graph reduction on firmer theoretical ground, characterising which coarsenings preserve the spectrum and the cut structure of the original graph [29] and asking how message passing should be redefined on a coarsened graph so that the propagated signal is provably preserved [30]. That question is directly relevant here,

since the decoder of this work propagates over exactly such a coarsened graph, and it is not one this thesis resolves.

What is contracted first, however, depends entirely on the cost assigned to a candidate merge, and here the relevant criterion comes from hierarchical clustering rather than from graph partitioning: Ward's linkage merges the pair whose union incurs the smallest increase in within-cluster sum of squares [10]. Chapter 5 shows that this choice, and not adaptivity as such, is what decides whether the method works.

Position of this work, and how the comparison is framed

Against this background the position of this thesis can be stated precisely, and what separates it from each of the families above is the same thing in every case: none of them can change the *shape* of the units it encodes. Fixed-rate codecs vary the fidelity of each frame but not the number of frames. Variable-frame-rate codecs vary the number of frames, but along the time axis alone, on speech, and guided by an external notion of information density. Structure-aware codecs vary what is stored in each cell of the time–frequency plane but not the shape of the cells [24]. Graph neural networks have reached audio only for discriminative tasks [17, 18], where reconstruction never has to be attempted.

What a graph adds to this picture is one operation the others do not have. Coarsening — contracting adjacent nodes into a super-node under a criterion — lets the units of the representation take the shape the signal suggests, and leaves a graph on which the same encoder and decoder continue to run. It is native to the substrate rather than bolted onto it, and it is the reason this work is built on a graph at all. The question it makes askable, and the one this thesis takes up, is whether a partition produced that way yields a representation of music that is efficient in its own terms, and whether it can be inverted back into audio at all.

That question had not been asked before in this form, which is the sense in which the contribution is a first step rather than an increment. The edge sets explored here are a small part of what the substrate permits, and the measurements in Chapter 5 bear on the mechanism rather than on the limits of the approach.

One methodological consequence follows, and it is stated here rather than left for the reader to discover. The *published* figures of the systems reviewed above are reported on axes this thesis cannot use: bitrates presuppose a quantiser, and the waveform-domain metrics they report are meaningless for a model that reconstructs magnitude alone and recovers phase by iterative resynthesis [9]. No number quoted in this chapter is therefore ever used as a baseline; those

figures are context and motivation only.

The absent quantiser is a decision rather than an omission, and the reasons are worth giving because they shape what the following chapters can and cannot claim. Residual quantisation is a rate mechanism in its own right, trained alongside the encoder and carrying its own failure mode — the codebook collapse that limits how many stages can be used productively, and that the Descript codec was largely built to address [3]. Placing it in series with an adaptive coarsener would put two mechanisms on the same axis, so that a change in rate could no longer be attributed to the partition. The experimental argument of this thesis is a single-variable comparison, and a quantiser is the one component that would spoil it. The variational bottleneck supplies what the quantiser would otherwise have been needed for: a rate in nats, measurable per dimension and differentiable, obtained without training a codebook. And keeping the code continuous is what makes the representation itself examinable, since the analysis in Chapter 5 traverses the latent space rather than inspecting a set of entries.

The price is real and is paid deliberately. Without a quantiser there is no bitrate, so no comparison with the systems above can be conducted in transmitted bits; the comparisons that follow are conducted at matched token count, and the cost of transmitting the partition, which a fixed grid does not incur, is measured separately in Chapter 5 rather than assumed away.

What is used as an external reference is a codec run here, under conditions this thesis controls. EnCodec [2] was executed at five rates on the same 4096 test clips as every system compared in Chapter 5, and it enters the argument in two roles: as a check that the evaluation metric can order a codec by its own rate, which the reconstruction metrics of this work cannot, and as an external marker of the quality level reachable at a comparable token count. Both roles are bounded, and the bounds are stated where the results appear: the comparison holds at matched token count and never in transmitted bits, on a metric of the magnitude domain that is largely blind to the phase this model discards, and against a system twenty-two times larger trained on a different domain. It is a reference point, not a contest.

The comparison that carries the experimental argument remains internal and controlled: a content-blind pooling operator that shares the encoder, decoder, objective and bottleneck of the adaptive branch and differs from it in one respect only, namely the assignment of graph nodes to tokens. It measures less than a comparison against the state of the art would, and it measures it without confounds.

3 Methodology

This chapter describes the model. The presentation follows the path of the signal: from the waveform to the time–frequency graph (Section 3.1), through the encoder and the adaptive coarsening that produces the tokens (Section 3.2), back to a spectrogram and to audio (Section 3.3), under the objective that trains the whole (Section 3.4). Section 3.5 collects the hyperparameters and the size of each component, and states the two properties of the implementation that constrain how the results of the following chapters may be read.

Figure 3.1 gives the pipeline in full. Two features of it are worth noticing before the details. First, the only stage whose output size depends on the content of the signal is the coarsening: everything upstream of it is a fixed function of the clip duration, and everything downstream is driven by the partition it produces. Second, the graph is built once, at full resolution, and is never rebuilt; the coarsener contracts it, and the decoder expands along the same contraction.

3.1 Preprocessing

The dataset

All experiments use NSynth [4], a corpus of individually recorded musical notes. Each clip is four seconds long, monophonic, sampled at 16 kHz, and contains a single note played by a single instrument with a well-defined attack, sustain and decay. The training split contains approximately 289 000 clips. The splits distributed with the corpus are used as they are, with-

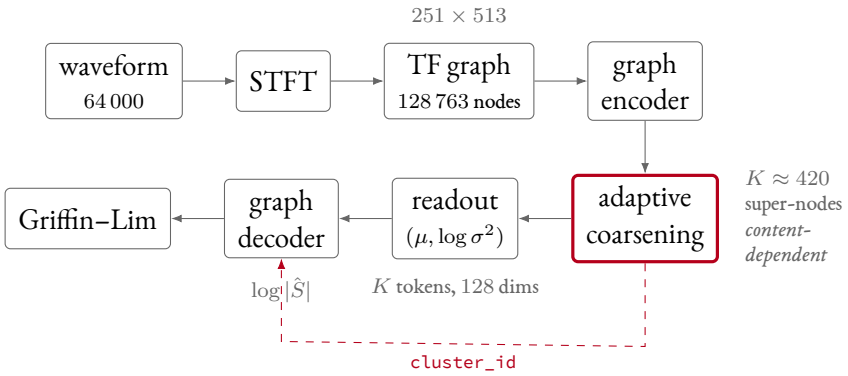


Figure 3.1: The complete pipeline. Solid arrows carry node or token features; the dashed arrow carries the partition itself, which the decoder needs in order to place each token back on the graph. The coarsening stage, highlighted, is the only one whose output size depends on the content of the signal rather than on the duration of the clip.

out resampling or reassignment: the training and validation directories are read directly, and the test split played no part in training or in model selection: it is used only for the evaluations of Chapter 5 — the distributional comparison, the listening test and the analysis of the latent space.

Two properties of the corpus matter for what follows. The first is that its time–frequency plane is mostly empty: measured on the full validation split, 62.2% of the bins fall below the silence threshold defined in Section 3.4. The second is that this proportion varies enormously across instrument families, from a plucked guitar whose energy has collapsed within a fraction of the window to a sustained voice that fills it. A corpus of isolated notes therefore presents the non-uniformity of information in an almost idealised form, which is what makes it a good testbed for an adaptive partition and, at the same time, a limited one: NSynth is not polyphonic music, and Chapter 6 returns to that limit.

A balanced subset of 550 training and 100 validation clips, obtained by sampling a fixed number of clips per instrument family, is used for the ablation campaign of Chapter 4, where the number of configurations to be compared makes the full corpus impractical. It is deliberately not representative: it is balanced by family while NSynth is not, and its active fraction is correspondingly higher. Every table derived from it is labelled as such.

From waveform to spectrogram

Audio is loaded as mono at 16 kHz, peak-normalised to $[-1, 1]$, and fixed to exactly 64 000 samples; longer clips are truncated and shorter ones zero-padded, though on NSynth neither

operation applies. The short-time Fourier transform uses a Hann window of 1024 samples with a hop of 256 and centred framing, giving a complex spectrogram $S \in \mathbb{C}^{T \times F}$ with $T = 251$ frames and $F = 513$ frequency bins.

The window length deserves a word, because it is not a neutral choice. At $n_{\text{fft}} = 512$ a frequency bin spans roughly 31 Hz, so the partials of a bass note fall within about two bins and are unresolvable before any model has seen them; doubling the window separates them, and it also lowers the ceiling imposed by phase resynthesis, discussed in Section 3.3.

The transform is applied to the clip as a whole rather than to a sequence of overlapping windows. A patch-based scheme with fifty per cent overlap would present roughly twice its input to the network in redundant form, and the redundancy would then have to be undone by the reconstruction. Each clip is therefore a single graph.

The time–frequency graph

Each bin of the spectrogram becomes a node. The feature of node i is a single scalar, the log-magnitude

$$x_i \;=\; \log\bigl(|S_{t_i,f_i}| + \varepsilon\bigr), \qquad \varepsilon = 10^{-8}, \tag{3.1}$$

and the phase is not part of the graph at all. The reason is structural rather than empirical: the coarsener aggregates the features of the nodes it merges by averaging them, and the average of two phases is not a phase. Whatever is carried through the bottleneck must be a quantity that survives averaging. Phase is instead reconstructed at synthesis time, at a cost quantified in Section 3.3.

Edges connect each node to its four lattice neighbours in time and frequency,

$$E_{\text{grid}} \;=\; \bigl\{(t,f) \leftrightarrow (t{+}1,f)\bigr\} \;\cup\; \bigl\{(t,f) \leftrightarrow (t,f{+}1)\bigr\}, \tag{3.2}$$

which on this grid yields $513\,524$ directed edges. Each edge carries a weight combining feature similarity and spatial proximity,

$$w_{ij} \;=\; \exp\!\left(-\frac{\|x_i - x_j\|^2}{\sigma^2}\right) \cdot \exp\bigl(-\alpha \|p_i - p_j\|\bigr), \tag{3.3}$$

with p_i the grid coordinates of node i and $\sigma = \alpha = 1$. The first factor is a Gaussian kernel on the log-magnitudes, the second an exponential decay in grid distance; their product penalises edges that are far apart in both senses at once.

Three further edge families are implemented and are disabled in every experiment reported in this thesis: harmonic edges connecting a candidate fundamental to its overtones, k-nearest-neighbour edges in a composite feature space, and distance-thresholded edges. The harmonic family is the one that matters argumentatively, since a graph carrying only grid edges performs message passing on a regular lattice and is, in that respect, a convolution; the ablation in Chapter 4 measures what those edges contribute and finds the answer to be negative, which is why they are off here.

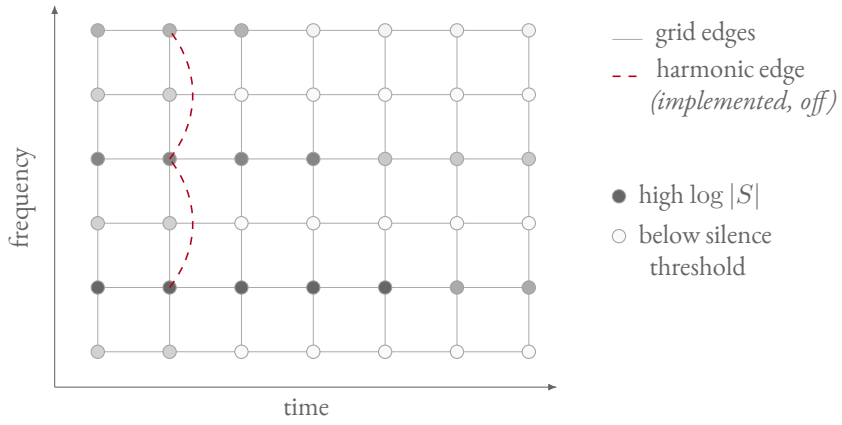


Figure 3.2: The time–frequency graph, drawn schematically on a 7×6 patch of what in the model is a 251×513 grid. Every bin is a node carrying its log-magnitude as its single feature; solid edges join the four lattice neighbours in time and frequency. The dashed edges connect a fundamental to its partials: they are implemented and are disabled in every experiment reported here, for the reasons given in Chapter 4. Note what the grid-only graph amounts to — message passing on a regular lattice is a convolution, so it is the non-grid edges that would distinguish the encoder from a convolutional one.

3.2 The encoder and the adaptive coarsening

Graph convolution

The encoder contextualises every node with its neighbourhood using Chebyshev spectral filters [6]. Let A be the weighted adjacency matrix, D its degree matrix, and

$$\tilde{L} \;=\; I - D^{-1/2} A D^{-1/2} \tag{3.4}$$

the symmetrically normalised Laplacian, whose spectrum lies in $[0, \lambda_{\max}]$ with $\lambda_{\max} \leq 2$. Rescaling it to the canonical interval of the Chebyshev polynomials, $\hat{L} = (2/\lambda_{\max})\tilde{L} - I$,

the recurrence

$$X^{(0)} = X, \qquad X^{(1)} = \hat{L}X, \qquad X^{(k+1)} = 2\hat{L}X^{(k)} - X^{(k-1)} \tag{3.5}$$

generates a basis in which a filter with K terms, a polynomial of degree $K-1$ in $\hat{L}$, is a learned combination

$$Y \;=\; \sum_{k=0}^{K-1} X^{(k)}\Theta^{(k)}, \qquad \Theta^{(k)} \in \mathbb{R}^{C_{\text{in}} \times C_{\text{out}}}. \tag{3.6}$$

The construction never forms $\tilde{L}$ explicitly and never computes an eigendecomposition: each term costs one sparse product, so the layer is linear in the number of edges, which on a graph of this size is the difference between a feasible model and an impossible one. The filter is also localised, since $X^{(k)}$ mixes only nodes within k hops, so a filter with K terms reaches $K-1$ hops. With $K = 3$ each layer reaches two hops, and six layers give a receptive field of twelve hops of the lattice, enough to place every bin in the context of its neighbours in both time and frequency without the smoothing that deeper stacks produce.

Each of the six layers is a residual block

$$H_{\ell+1} \;=\; H_\ell + \phi\big(\mathrm{GraphNorm}(\mathrm{ChebConv}_\ell(H_\ell))\big), \tag{3.7}$$

with ϕ the SiLU non-linearity [31] and a hidden width of 128. The normalisation is GraphNorm [32] rather than batch normalisation, and the weights are initialised by a Xavier scheme [33] scaled by $1/\sqrt{K+1}$, a slightly conservative compensation for the K polynomial terms summed in (3.6), with an additional $1/\sqrt{2L}$ factor on the last layer of the stack.

Node scoring

Before the graph is contracted, each node receives a scalar importance score. The scorer sees the encoder embedding h_i together with two purely structural quantities, the log-degree and the mean weight of the incident edges,

$$u_i \;=\; \big[\, h_i,\;\; \log\max(d_i, 1),\;\; \bar{w}_i \,\big], \qquad \sigma_i \;=\; \mathrm{MLP}(u_i) \in \mathbb{R}, \tag{3.8}$$

and the score plays two roles downstream: it weights the pooling that produces each super-node feature, and it defines a preserved set of nodes that may receive merges but never disappear into another node.

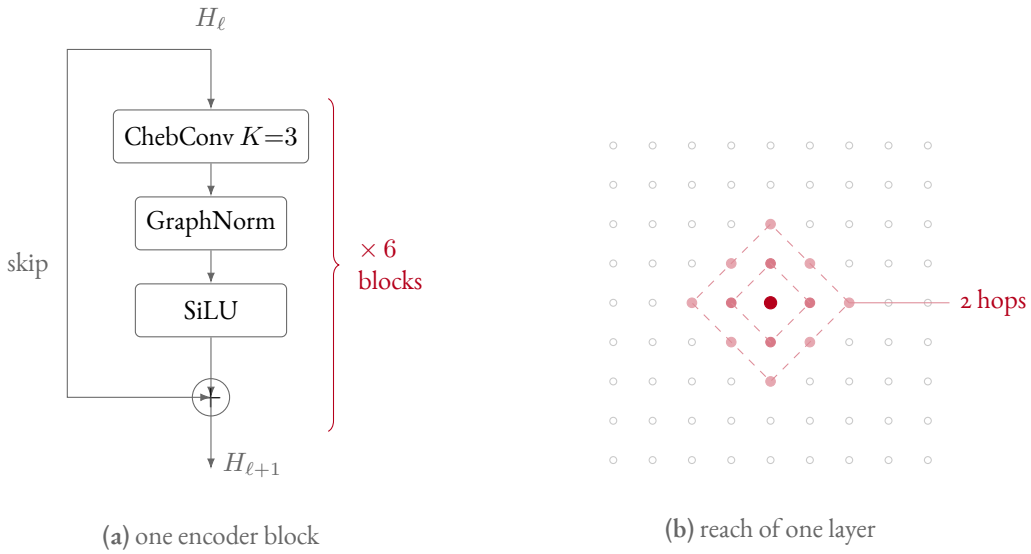


Figure 3.3: The encoder. **(a)** One residual block, repeated six times. **(b)** What the polynomial order controls: a filter with K terms mixes a node only with the nodes within $K-1$ hops of it, and on the four-neighbour lattice those form a diamond. Six blocks with $K=3$ give each node a reach of twelve hops, enough to span a neighbourhood in both time and frequency without the over-smoothing that deeper stacks produce.

Adaptive coarsening

This is where the token sequence acquires its length, and it is the component on which the results of Chapter 5 turn.

The coarsener maintains a cluster assignment $c \in \{0, \dots, K-1\}^N$, initialised to the identity, and reduces it in rounds. Each round builds the current super-graph by pooling node features within clusters, using the scores as weights,

$$\bar{x}_k = \frac{\sum_{i:c_i=k} \sigma_i \, x_i}{\sum_{i:c_i=k} \sigma_i}, \qquad W_k = \sum_{i:c_i=k} \sigma_i, \tag{3.9}$$

collapses the edges onto the cluster space and removes the self-loops. Features are then L_2-normalised, which bounds every pairwise squared distance to $[0, 4]$ and lets a single fixed kernel width behave consistently regardless of the scale of the embeddings.

Each candidate merge is scored, all candidates are sorted, and a matching is extracted greedily so that many disjoint pairs are contracted in one round; the strategy is the heavy-edge matching of multilevel graph partitioning [7], adapted to operate on learned features rather than on given edge weights. The number of rounds is not fixed in advance. Merging stops when no candidate

offers a benefit above a threshold, or when the node count falls below a ceiling $K_{\max}$; which of the two regimes applies is a configuration choice with consequences discussed in Section 3.5.

The merge cost

What is contracted, and therefore what the tokens are, depends entirely on the cost assigned to a candidate pair. Two criteria are compared throughout this thesis.

The first minimises the *graph total variation* of the node features,

$$\mathrm{GTV}(X;\mathcal{G}) \;=\; \sum_{(i,j)\in E} w_{ij}\|x_i - x_j\|_2, \tag{3.10}$$

a standard smoothness functional for signals on graphs [26]. For a candidate pair (a, b) the criterion computes the variation contributed by all edges incident to either endpoint, the variation that would be contributed by the merged node, and takes the difference

$$\Delta_{ab} \;=\; \mathrm{old}_{ab} - \mathrm{new}_{ab}, \tag{3.11}$$

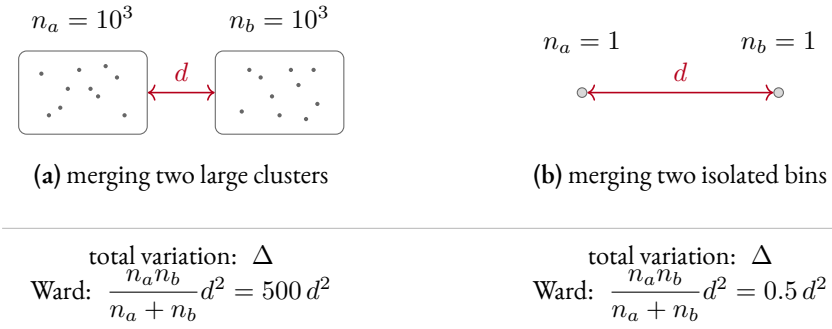
preferring the pair whose contraction lowers the total variation most.

The second is Ward's linkage [10], the increase in within-cluster sum of squares caused by a merge,

$$\mathrm{cost}_{ab} \;=\; \frac{n_a\, n_b}{n_a + n_b}\,\big\|\mu_a - \mu_b\big\|_2^2, \tag{3.12}$$

where n_a and n_b are the sizes of the two clusters and μ_a, μ_b their means.

The difference between the two is a single factor, and it is the point of the whole construction. The variation criterion (3.11) has no term in the cluster sizes: absorbing a cluster of a thousand nodes into another costs exactly what merging two isolated bins costs, so a greedy matching has no reason to prevent a few clusters from consuming the plane. Ward's cost (3.12) weights the distance between the means by the harmonic size term $n_a n_b/(n_a + n_b)$, and what it thereby measures is precisely the squared error that the subsequent mean-pooling will commit on the merged cluster. The criterion, in other words, is aligned with the loss that the reconstruction will pay. Chapter 5 shows that this alignment, and not adaptivity as such, is what determines whether the method works at all.



(a) merging two large clusters

(b) merging two isolated bins

Figure 3.4: Why the size term decides the partition. Both panels show a candidate merge between two clusters whose feature means are a distance d apart. A criterion based on the reduction of total variation sees only that distance, so it prices the two merges identically: absorbing a thousand-node cluster into another costs what merging two isolated bins costs, and a greedy matching has no reason to stop a few clusters from consuming the plane. Ward's linkage weights the distance by $n_a n_b/(n_a + n_b)$, which is exactly the squared error that the subsequent mean-pooling will commit, and separates the two cases by three orders of magnitude.

A structural limitation

The coarsening rounds run under `torch.no_grad()`. No gradient reaches the decision of which nodes to merge: the partition is *guided* by the encoder, since the costs are computed on its embeddings, but it is not *learned*. The encoder and the scorer receive gradient only through the final pooling, which teaches them what to put inside a token and how to weight the nodes within a cluster, never how to form the clusters. This is a deliberate consequence of using a discrete greedy matching rather than a differentiable relaxation, it is the reason a more expressive front end can make the partition worse rather than better, and it is stated here because several results in Chapters 4 and 5 are only intelligible in its light.

Readout

The surviving super-nodes become tokens. A two-layer MLP is applied node-wise,

$$z_k \;=\; W_2\,\phi(W_1 \bar{x}_k + b_1) + b_2 \;\in\; \mathbb{R}^{128}, \tag{3.13}$$

which commutes with any permutation of the node indices, as it must: the tokens are a set indexed by the partition, not an ordered sequence. There is no graph-level pooling, since every super-node becomes its own token rather than contributing to a single vector for the whole

clip.

The readout is variational. It projects to a mean and a log-variance and the divergence of the resulting posterior from the standard normal prior,

$$\mathcal{L}_{\mathrm{KL}} \;=\; -\tfrac{1}{2}\sum_{d}\bigl(1+\log\sigma_d^2-\mu_d^2-\sigma_d^2\bigr), \tag{3.14}$$

is an information rate in nats, adjustable through its weight β [8]. This provides the second rate axis used in Chapter 5, and it does so without a codebook: residual vector quantisation is implemented in the codebase but is disabled in every experiment reported here, so a token is a continuous vector and no bitrate in kilobits per second is claimed anywhere in this thesis. At evaluation time the readout returns the mean rather than a sample, which makes the representation deterministic.

3.3 Decoder

The decoder inverts the path in three stages, and the second of them is the one that makes the whole construction possible.

A first stack of graph convolutions operates on the coarsened graph, on the K tokens and the edges inherited from the contraction, in blocks identical in form to (3.7). Self-loops are added beforehand so that no node has zero degree, which would make the normalised Laplacian of (3.4) singular.

The tokens are then scattered back onto the original grid using the cluster map produced by the coarsener,

$$H_i^{f} \;=\; Z_{c_i}, \qquad i=1,\dots,N, \tag{3.15}$$

or in matrix form $H^{f} = S^{\top}Z$ with $S \in \{0,1\}^{K\times N}$ the cluster-membership matrix. At this point every node inside a cluster carries exactly the same vector: the expansion restores the resolution but not the detail. It is worth being explicit that this step consumes the partition itself. The decoder cannot place a token without knowing which nodes it summarises, so the map c is as necessary to reconstruction as the token values are, a fact whose consequences for the accounting of rate are taken up in Chapter 6.

A second stack of graph convolutions, on the original full-resolution edges, then restores the variation within each cluster; without it, nodes belonging to the same super-node would remain identical to one another all the way to the output. A single linear layer maps each hidden

vector to one channel, the predicted log-magnitude.

Reconstruction to audio exponentiates the predicted log-magnitude and recovers the phase with the Griffin–Lim algorithm [9], run for 32 iterations. This places a ceiling on what the model can achieve that has nothing to do with the model: even a perfect magnitude, resynthesised this way, does not return the original waveform. Controlled round trips put that ceiling at roughly 0.06 in the mel distance of Section 3.4, and they also show that scale-invariant signal-to-noise ratio is irrecoverable for a magnitude-only codec, which is why no waveform-domain metric is reported as a result anywhere in this thesis.

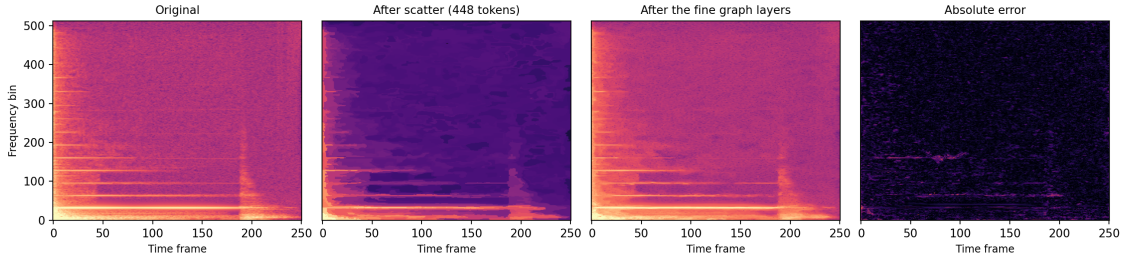


Figure 3.5: The decoder on one clip of the test split, with the intermediate stage made visible. After the scatter of equation (3.15) every node of a cluster carries the same vector, and projecting that state through the output head gives the second panel: the resolution of the original grid has been restored but none of the detail, and the partition is directly legible as flat regions. The graph layers that follow recover the variation within each cluster, giving the third panel. The intermediate panel is shown on its own colour scale, since the values at that depth are not yet calibrated in level and the shared scale would hide the structure; the other three share one scale.

3.4 Loss

The objective has three active terms,

$$\mathcal{L} \;=\; \mathcal{L}_{\text{node}} \;+\; \lambda_{\text{gtv}}\,\mathcal{L}_{\text{gtv}} \;+\; \beta\,\mathcal{L}_{\text{KL}}, \tag{3.16}$$

with $\lambda_{\text{gtv}} = 0.1$ and $\beta = 10^{-3}$ in every experiment reported here.

The reconstruction term is an L_1 distance between the predicted and the target log-magnitude, taken per node, but weighted so that the note dominates the gradient:

$$\mathcal{L}_{\text{node}} \;=\; \frac{1}{N}\sum_{i=1}^{N} \omega_i \left|\hat{x}_i - x_i\right|, \qquad \omega_i \;=\; \gamma + (1-\gamma)\, s\!\left(\frac{x_i - \theta}{\tau}\right), \tag{3.17}$$

where s is the logistic function, $\theta = -7$ is the silence threshold, $\tau = 1$ its temperature, and

$\gamma = 0.2$ the weight retained on silent bins. The need for this weighting follows from the statistics of the corpus given in Section 3.1: on a plane that is more than sixty per cent silence, the cheapest constant a model can emit is the noise floor, and an unweighted mean would reward it for doing so. The same threshold θ defines the active and silent partitions used by the diagnostic metrics of Chapter 4, and those metrics apply it to the *target* and never to the prediction, so that a model cannot improve its own score by predicting silence.

The second term is the graph total variation of (3.10) evaluated on the reconstruction, normalised by the number of unique undirected edges so that its scale does not depend on the size or the connectivity of the graph. It acts as a smoothness prior on the output. The third is the divergence of (3.14), which is simultaneously a regulariser and the measured information rate of the representation.

Evaluation uses the mel distance, an L_1 distance between the log-mel spectrograms of the original and the reconstruction, as the primary metric, reported alongside the same node-wise L_1 split into its active and silent parts.

3.5 Parameters discussion

Model size

The complete system has $654\,850$ parameters, distributed as in Table 3.1.

Table 3.1: Parameter count by component, with the variational readout enabled.

Component	Configuration	Parameters
Encoder	6 ChebConv blocks, $K = 3$, width 128	249 472
Scorer	per-node MLP over $[h_i, \log d_i, \bar{w}_i]$	8 449
Readout	two-layer MLP to $(\mu, \log \sigma^2)$, 128 dims	98 816
Decoder	$3 + 3$ ChebConv blocks and a linear head	298 113
Total		**654 850**

For comparison, RAVE [5] has 17.6 million parameters, a factor of 27. This is a design commitment rather than a limitation accepted under duress, and it should be read as one: the decoder, which carries the largest share at 45%, must reconstruct $128\,763$ nodes from roughly 420 tokens, and it does so with fewer parameters than a single layer of most published codecs.

Optimisation

Training uses AdamW [34] with a learning rate of 10^{-3}, weight decay 10^{-5}, 100 warmup steps followed by a cosine decay, and gradient-norm clipping at 1.0. Batches contain four clips. Runs terminate by early stopping on the total validation loss, and every figure reported in this thesis is taken at the best checkpoint rather than at the last epoch.

One convention needs stating because it affects how the training curves should be read: an epoch here is a fixed number of optimisation steps over clips drawn from the whole training split, not a complete pass over the corpus. With 289 000 clips available, no run in this thesis has seen the training set in its entirety more than a few times.

Two properties that bound the results

Two features of the implementation determine how the numbers in the following chapters may be read, and both are stated here so that no result has to carry the explanation with it.

The runs are not seeded, and the resulting noise has been measured. Training two configurationally identical arms and comparing them gives a difference of 4.5% in mel distance, which is the reproducibility of the whole pipeline — initialisation, data order and the nondeterministic reductions of the coarsener taken together. That figure is used throughout as a floor: a difference smaller than it is not read as an effect anywhere in this thesis. Seeding the runs would have narrowed it, at the cost of leaving the variability unmeasured; measuring it once and applying it as a threshold is the more conservative arrangement, and it is what makes a statement such as *fourteen times the noise floor* meaningful rather than rhetorical.

The validation figures are selected on the split they are reported on, and the consequence was checked rather than assumed. Every arm is stopped early on the total validation loss and every figure is taken at its best checkpoint, so each reported value is the minimum of several noisy evaluations on the same set — optimistically biased by construction, and by an amount that differs between arms, since they ran for different numbers of epochs. The bias was therefore measured directly: each best checkpoint was re-evaluated on the *test* split, which no run had seen and on which no selection had taken place. The ordering of the arms is unchanged, and the gap between the adaptive and the fixed partition at a matched token count moves from 1.38 to 1.40 times. The selection bias is real and it is small, and the results of Chapter 5 that rest on the held-out split — the distributional evaluation, the listening test, the analysis of the representation — are free of it by construction.

4

Experiments

This chapter describes the experimental campaign: what was measured, how, and what came out. Chapter 5 is reserved for the results the thesis rests on, and the division is one of function rather than of importance: an ablation that isolates a component and an estimate of what the partition costs to transmit both serve to make the results of the next chapter readable, and are reported here beside the designs that produced them.

Section 4.1 isolates the components of the model against one another, on a small balanced subset where many configurations can be swept cheaply, and reports what each removal costs. Section 4.2 then addresses a harder problem: the metrics used throughout the ablations cannot arbitrate a comparison against a system built on different principles, and a different kind of measurement is required. It also counts what it costs to transmit the partition, which is the accounting the token count omits. Section 4.3 describes a blind listening test whose purpose is to validate the chosen metric against human hearing. Section 4.4 stands apart from the other three, in that it does not compare systems at all but examines what the bottleneck of the winning model came to contain.

4.1 Ablations

Protocol

Every experiment in this section uses the balanced subset described in Section 3.1, trains for 60 epochs, and caps the node count at $K_{\max} = 512$. The cap is not incidental. With the ceiling active the coarsener is forced to merge until it reaches it, so the stopping threshold never gates and every arm lands at essentially the same token count; what differs between arms is then only *which* merges were chosen, never how many. A comparison of merge criteria at unequal rates would confound the two.

Two rate axes are used throughout this thesis and both are defined here. The first is the token count, which is natural but coarse, since a token is a continuous vector and nothing in this work has been quantised. The second is supplied by the variational bottleneck: the Kullback–Leibler divergence to the prior is an information rate in nats, and the total for a clip is the mean divergence per token multiplied by the number of tokens. The two are not independent by construction, but neither restates the other — the number of tokens is decided by the coarsener, while the divergence per token is decided by what the encoder chooses to transmit — so an ordering preserved on both is stronger evidence than an ordering on either.

Harmonic edges are disabled except where they are the object of the ablation, and the merge bonuses that reward perceptually or structurally motivated contractions are set to zero, since their scales are not comparable across cost conventions. The noise floor established in Section 3.5 applies throughout: a difference below 4.5% is not read as an effect.

What the graph convolution contributes

The premise of this work is that a graph is the right substrate for an adaptive partition, and whether the *propagation* over that graph earns its cost is a separate question that twenty-five groups of experiments had left unmeasured. It was checked before the premise was built on any further. Reducing the Chebyshev filter of (3.6) to its order-zero term leaves a per-node linear map with no neighbour aggregation, everything else in the model unchanged; because that also removes 55% of the parameters, a third arm restores the count by adding depth. **Removing propagation raises the mel distance by 65%, fourteen times the noise floor, and the capacity-matched arm reproduces the degraded result rather than the intact one.** Contextualising a node with its neighbours is therefore doing substantial work, and the degradation is not a matter of network size.

Two properties bound the claim. The order K is shared between encoder and decoder, so the ablation removes propagation from the whole model rather than from the encoder alone. And **the graph does not disappear**: the coarsener still matches along real edges and the decoder still scatters along the cluster map, so what is measured is the value of contextualising a node with its neighbours, not the value of the graph as such.

The merge criterion

The criterion governing which pairs are contracted is varied along two axes at once, since the proposal changed both together: the cost function, total variation against Ward's linkage, and the feature space in which distances are measured, the encoder embeddings against the raw log-magnitude. The full 2×2 separates the contribution of each.

The second axis deserves an explanation, because it is the one that carries a risk for the thesis. Computing the merge cost on the raw log-magnitude means computing the partition *without the encoder*: the graph convolution still produces the values that are pooled into tokens, but it no longer influences where the cluster boundaries fall. If that arm were to match the one that uses embeddings, it would follow that the network contributes to the representation but not to the partition. The experiment was run in that knowledge, and Chapter 5 reports what it found.

The 2×2 separates the cost function from the space in which distances are measured. The two axes do not behave alike.

Table 4.1: Merge criterion, at a rate pinned by the node ceiling. The cost function is varied against the feature space in which distances are measured.

Cost	Feature space	Tokens	Mel ↓	L_1 act. ↓
Total variation	GNN embeddings	418.6	1.1520	0.7763
Total variation	Log-magnitude	461.7	1.1759	0.8084
Ward	**GNN embeddings**	**431.5**	**0.8125**	**0.5742**
Ward	Log-magnitude	413.7	0.8186	0.5770

The two axes behave completely differently. Changing the cost function improves the mel distance by 29%; changing the space in which distances are measured moves it by less than the noise floor. Both readings matter, and the second is uncomfortable.

The size term is what decides the partition. Without it, absorbing a thousand-node cluster into another is priced like merging two isolated bins, and greedy matching has no reason to

prevent a few clusters from consuming the plane.

The feature space, on the other hand, does not decide anything. The arm that computes merge costs on the raw log-magnitude reaches the same quality as the one that computes them on the encoder embeddings, which means **the partition does not need the graph convolution**. Read together with the propagation ablation of Section 4.1, the two results locate the contribution of the network precisely: it is in the encoding, worth 65%, and not in the partition, worth nothing measurable. The explanation is structural and was stated in Section 3.2: the coarsening runs without gradients, so no signal ever asks the encoder to produce representations suited to deciding *where* to merge. It is asked only what to put inside a token.

Harmonic edges

Edges joining a fundamental to its partials are the component that would most obviously distinguish this encoder from a convolutional one, since a graph carrying only grid edges is a lattice and message passing on a lattice is a convolution. Their affinity is Gaussian in a width parameterised in cents and learned during training, and they are compared against the time–frequency grid alone at two node ceilings.

Table 4.2: Harmonic edges against the time–frequency grid alone, at two node ceilings. At 8192 the grid-only arm wins having trained for less than half as long.

Ceiling	Harmonic edges	Tokens	Mel ↓	Epochs
128	yes	121.2	2.3840	25
128	no	109.4	**1.6345**	25
8192	yes	6041.6	1.5084	40
8192	no	6446.0	**1.0154**	16

The result is negative and reproduces at both ceilings. The mechanism is measurable rather than conjectural. Between 37 and 88 per cent of merges follow a harmonic edge, so the edges are used, and using them merges bins at different frequencies, which flattens the spectral envelope and misplaces the pitch; the grid merges locally and the pitch survives. What fails is therefore the harmonic prior specifically and not the graph, since propagation is worth 65% on the grid alone.

One quantity from these runs is worth keeping for a different reason. The learned width converges from an initialisation of 50 cents to approximately 28 in two independent configurations, which is an estimate of how far the partials of this corpus depart from exact harmonicity

and the only parameter in the model that admits a physical reading. It is reported as an observation about the parameter, not as a quality comparison.

Collapsing silence

If silence is cheap to describe, the most direct way to exploit that is to say so in advance: detect the silent region and collapse it into a small fixed number of super-nodes before the coarsener runs, freeing the budget for the note. This ablation tests whether structure of that kind is better imposed or left to be found. The experiment sweeps that number over seven values, from a single super-node for the whole silent region up to 256.

The intervention has a side effect that the experiment was designed to expose. Sealing the silent nodes off from the merging process partitions the graph: the active region is no longer connected through the silence, and it breaks into components that can never be merged with one another. Measuring the number of connected components of the active region, alongside the reconstruction error split into its active and silent parts, is what distinguishes *the budget was reallocated badly* from *the partition was fragmented*.

This is the direct test of the hypothesis the thesis started from.

Table 4.3: Collapsing the silent region into K super-nodes before coarsening. No configuration reaches the model that gives silence no special treatment, and all of them spend more tokens.

Silent subgraph	Tokens	Mel ↓	L_1 act. ↓	L_1 sil. ↓
No collapse (Ward)	**431.5**	**0.8125**	**0.5742**	**0.4415**
$K = 1$	545.0	1.1342	0.6674	0.9348
$K = 16$	555.0	1.1773	0.7168	0.8321
$K = 64$	556.4	1.0249	0.6653	0.7009

The intervention predicts the opposite of this table, and the way it fails is more informative than the failure. Sealing off the silent nodes fragments the active region into components that can no longer be merged with one another: measured over 120 clips, the active region breaks into 392 connected components on average, with a median of 222. The token count then settles *above* the imposed ceiling rather than below it, roughly 550 against 512, and is flat from the first epoch. The partition degrades in shape before it degrades in size, which is why three times the budget on the signal yields a worse reconstruction of it.

What fails is not the idea that silence is cheap — Section 5.2 shows the trained model arriving at a codeword for silence on its own — but the mechanism of reserving budget by *partitioning*

the graph before the coarsener can act. Structure of this kind is better induced than imposed. What the freed budget would buy, if it were freed without fragmenting the domain, remains untested, and at high rates the same intervention does pay.

The rate–distortion axis of the bottleneck

The weight β of the variational term is swept over $\{0, 10^{-3}, 10^{-2}, 10^{-1}\}$ at a fixed node ceiling. Since the ceiling pins the token count, the only quantity that varies is how much information each token is permitted to carry, which makes the sweep a rate–distortion curve on the information axis alone, uncontaminated by changes in sequence length. Its purpose is to establish that $\beta = 10^{-3}$ is a chosen operating point rather than an arbitrary one, and that the divergence is a governable rate.

It is. Across the sweep the divergence falls by more than an order of magnitude, from 20.88 nats at $\beta = 10^{-3}$ to 0.81 at 10^{-1}, while the token count moves by 7% and the reconstruction degrades monotonically with the divergence. The bottleneck behaves as a rate control rather than as an incidental regulariser, which is what licenses the second rate axis to be used as one.

4.2 Measuring quality, and counting the partition

Why the reconstruction metrics are not enough

Every number in Section 4.1 is a mel distance or a node-wise L_1 error. Within the family of models trained here those metrics are trustworthy: the five arms share their parameters, their transform, their phase reconstruction and their objective, and differ only in the coarsening, so a difference between them is attributable to the coarsening.

Outside that family they break down, and the failure is not subtle. Measured on the same clips, the log-mel distance ranks a GAAT model ahead of EnCodec at $24\,\mathrm{kbit\,s^{-1}}$, sixteen times its rate, while the scale-invariant signal-to-noise ratio ranks it behind by almost 40 decibels. Two banal explanations were excluded: correcting for the optimal temporal shift moves the log-mel distance by less than 0.005, and gating the frames below $-40\,\mathrm{dB}$ does not reverse the picture.

The explanation is that each metric measures a different object and this model keeps only one of them. The log-mel distance measures agreement in magnitude, which is precisely what the objective of Section 3.4 optimises and what a waveform codec sacrifices for perceptual plausibility. The signal-to-noise ratio measures agreement in waveform, which a magnitude-only

model discards entirely along with the phase. Neither is a statement about quality, and **no reconstruction metric defined in this work can arbitrate a comparison between architectures**. A measurement is needed that is the objective of neither contender.

Fréchet distance over a contrastive audio embedding

The measurement adopted here embeds audio with CLAP [35], a model trained to place a recording and its textual description close together in a shared space. The text side is not used; what matters is that the audio encoder was trained on human-written descriptions, hence on a notion of similarity that coincides with none of the losses involved. The checkpoint is `laion/clap-htsat-unfused`, producing a unit-norm vector of 512 numbers per clip. It is preferred to the VGGish encoder conventionally used with this family of metrics because VGGish, trained for classification on AudioSet, is a poor space for music [36].

Two quantities are computed in that space. The first is the Fréchet distance [37] between the set of original clips and the set of reconstructions: a Gaussian is fitted to each population and

$$\mathrm{FAD} \;=\; \|\mu_r - \mu_c\|^2 \;+\; \mathrm{tr}\big(\Sigma_r + \Sigma_c - 2(\Sigma_r\Sigma_c)^{1/2}\big) \tag{4.1}$$

measures how far apart they are in centre and in shape, the two terms that Figure 4.1 makes visible.

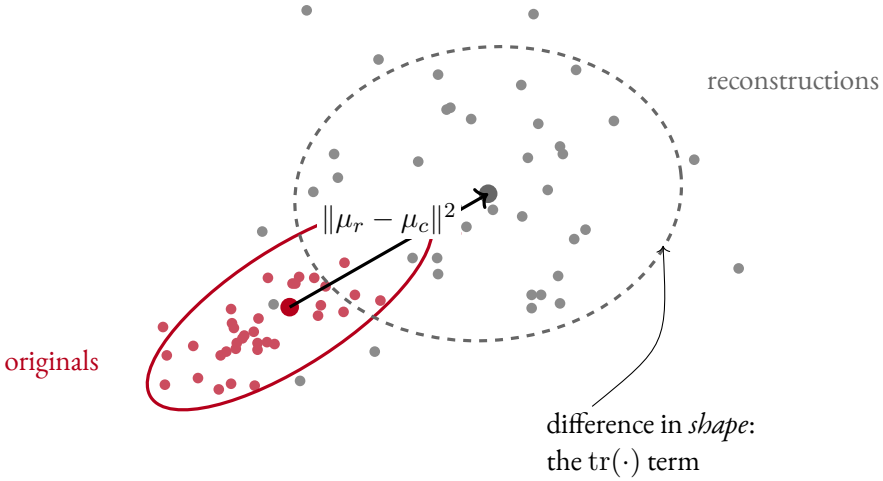


Figure 4.1: The two terms of (4.1). A Gaussian is fitted to each population of embeddings; the first term is the squared distance between their centres, the second compares their shapes. The figure is schematic: the points are drawn from two Gaussians and are not measured data.

That distance compares populations and is therefore blind to pairing: a system that recon-

structed every clip as some *other* clip of the corpus would score perfectly, as Figure 4.2 shows. The second quantity closes the gap by taking, for each clip, the cosine between the embedding of the original and that of its own reconstruction. It also buys resolution, because the variance across different clips is far larger than the difference between two coarsening schemes, and comparing clip by clip cancels it.

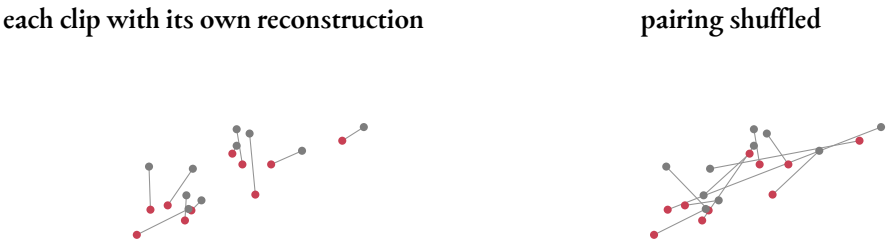


Figure 4.2: Why two quantities are computed rather than one. Both halves contain the same two populations and differ only in which reconstruction is associated with which original, so the Fréchet distance is identical across them while the paired cosine is not. Schematic, as above.

The evaluation runs on the **full test split of NSynth**, 4096 clips whose instruments are disjoint from those seen in training, and embeds eight systems: the originals as reference; a Griffin–Lim control receiving the *exact* magnitude of the original, which errs only in phase and therefore prices phase reconstruction alone, so that whatever a trained arm sits above it is what the coarsening has to account for; the five trained arms; and EnCodec [2] at $1.5\,\mathrm{kbit\,s^{-1}}$. The three uniform arms, spanning 196 to 2907 tokens, supply a rate axis along which the adaptive point can be located rather than compared against a single opponent.

Two details of the procedure are not free choices. NSynth clips are isolated notes while CLAP was trained on sound scenes, so the originals occupy a thin sliver of its 512-dimensional space — an effective dimensionality of 12.8 — and evaluating (4.1) in the full space would sum hundreds of axes carrying only sampling noise. A principal-component basis is therefore estimated *once, on the reference set* and applied unchanged to every system, which leaves the metric intact because the Fréchet distance is invariant under orthogonal changes of basis. And confidence intervals are computed on the *difference* between two systems over the same resampled clips: resampling duplicates clips, duplication shrinks the effective sample, and a smaller sample inflates a Fréchet distance by more than the quantity being measured — an inflation that largely cancels in a difference.

Two checks preceded any use of these numbers: the distances stabilise to the third decimal from 1024 clips onwards, and the metric was made to pass the test the reconstruction metrics

had failed, by evaluating EnCodec at five bitrates on the same clips. Both outcomes are reported in Chapter 5.

Counting the partition

Every rate quoted for this model so far counts tokens, and a token sequence is not sufficient to reconstruct: the decoder must also know which of the 128 763 grid nodes belongs to which token, and it cannot recompute that map because the map derives from encoder features the decoder never sees. In a transmitted codec the partition would have to be sent. The uniform baseline is what makes this measurable rather than merely worrying: its partition is a fixed grid defined by two integers known to both ends, so it costs exactly zero bits, and any advantage of the adaptive coarsener has to be paid out of that difference before it can be called an advantage in rate.

Three estimators are computed, all of them achievable rates and therefore upper bounds on the true minimum. The first assigns an independent label to each cell, and the second spends one binary decision per grid edge at the empirical boundary rate; both are deliberately crude, and serve as brackets. The third is the realistic one: a coder that walks the grid in raster order and predicts each cell from the neighbours already decoded above it and to its left, paying only when the prediction fails. Regions that are large and compact are cheap under it, scattered ones expensive. All three are validated against partitions of known cost, and the measurement covers 128 clips of the test split across the five trained arms.

Table 4.4: Cost of transmitting the partition, in kilobits per second, over 128 clips of the test split. The estimators for the uniform arms are italicised because they are hypothetical: a fixed grid is defined by two integers known to both ends, so its partition costs nothing and its total is the payload alone.

Arm	Tokens	Boundary frac.	Naive	Context	Payload	**Total**
Adaptive, Ward	424.2	0.173	280.7	19.6	1.89	**21.47**
Adaptive, TV	412.8	0.141	279.4	22.1	2.01	**24.11**
Uniform 26×12	430.0	0.059	*281.6*	*4.8*	2.46	**2.46**
Uniform 5×9	2907.0	0.155	*370.4*	*19.8*	20.51	**20.51**
Uniform 18×38	196.0	0.039	*245.1*	*2.9*	0.98	**0.98**

Under the realistic estimator the partition of the adaptive arm costs $19.6\,\mathrm{kbit\,s^{-1}}$ against 1.89 of payload: **ten times the tokens it describes**. The transmitted rate of the model is therefore $21.47\,\mathrm{kbit\,s^{-1}}$, not the 1.9 that the token count alone would suggest.

What this measurement bounds, and what it leaves alone, should be stated precisely, because the two are easy to conflate. It leaves the quality results alone: those are measured at a matched token count and say nothing about bits, so nothing in Chapter 5 is revised by this table. What it bounds is the word *efficient* in its rate sense. A representation that is cheaper by a factor of 6.7 in tokens is cheaper by between 1.0 and 1.6 in transmitted bits, and only the first of those is a claim this thesis makes.

The distinction is not a technicality, because the two numbers belong to different uses of the same object. Where the token sequence is consumed by a model downstream — the setting from which Chapter 1 takes its motivation — the partition is never transmitted at all: the encoder computes it and nothing crosses a channel, so the cost that matters is the length of the sequence and it is paid in tokens. The cost counted here is the price of the other use, in which the representation is sent somewhere, and that use is the one this work does not claim.

A structural property emerges from the same measurement and is worth recording independently: the number of connected components per region is exactly 1.00 on every arm. Each token corresponds to a single connected region of the grid, never to scattered fragments. This is a property of the coarsening rather than an accident, and it is what justifies the context coder used to estimate the cost.

Two currencies, both honest Placing the totals of Table 4.4 against the distances of Table 5.3 changes the shape of the result:

Arm	Total rate	FAD-CLAP ↓
Uniform 18×38	$0.98\,\mathrm{kbit\,s^{-1}}$	0.318
Uniform 26×12	$2.46\,\mathrm{kbit\,s^{-1}}$	0.289
Uniform 5×9	$20.51\,\mathrm{kbit\,s^{-1}}$	0.179
Adaptive, Ward	$21.47\,\mathrm{kbit\,s^{-1}}$	**0.156**

Ward costs 1.05 times the dense uniform arm and is 1.15 times better. Extrapolating the uniform curve, a fixed grid would reach the same distance at roughly $35\,\mathrm{kbit\,s^{-1}}$, a factor of 1.64 — but that lies outside the measured interval and is an indication, not a measurement. The advantage in transmitted bits is therefore **between** 1.0 **and** 1.6, not 6.7.

The two statements are both true and belong to different currencies:

> *In tokens*: the adaptive model uses 433 tokens against the 2907 of the uniform arm nearest to it in quality — 6.7 times fewer.

In transmitted bits: it costs 21.47 against $20.51\,\mathrm{kbit\,s^{-1}}$ — essentially the same rate, for a distance 15% better.

Which currency applies depends on what the representation is for. The token is the unit a downstream generative model consumes, and it is the unit in which the subject of this thesis — a latent representation — is denominated. The bit is the unit of a codec, and on that scale adaptive coarsening is not yet an advantage. This thesis is about the representation, so the statement in tokens is the principal one; the statement in bits is stated alongside it here rather than left for a reader to derive, because it is the objection that will come.

4.3 The subjective listening test

Every measurement described so far is a proxy. The Fréchet distance and the paired cosine of Section 4.2 were adopted because the embedding they are computed in was trained on human-written descriptions of sound, which is an argument about provenance rather than evidence about perception. The only way to establish that any of these numbers track what a listener hears is to ask listeners, and to ask them in a way that admits a negative answer.

Stimuli and conditions

Seven clips were drawn from the test split of NSynth, from five of the ten instrument families it contains: two *bass*, two *organ*, and one each of *brass*, *flute* and *vocal*. Guitar, keyboard, mallet, reed and string are absent, which bounds the generalisation of the result and is stated again with the result. The selection is not arbitrary in one respect that matters: *brass* and *vocal* are the two families whose notes sustain to the end of the four-second window rather than decaying into silence, so they exercise the coarsener in the regime where it has no empty region to absorb.

Five conditions were rendered for each clip, giving thirty-five judgements per participant: the unprocessed audio as a **hidden reference**; the adaptive Ward model at roughly 433 tokens; the uniform 26×12 grid at 430 tokens, which is the rate-matched control; the adaptive total-variation model at roughly 418 tokens; and EnCodec at $1.5\,\mathrm{kbit\,s^{-1}}$. The Griffin–Lim ceiling exists on disk and was deliberately excluded: that comparison is already settled by the metrics of Section 4.2, and every additional condition costs each participant seven more listening passes.

Three controls, and why each is necessary

Loudness. All thirty-five tracks were normalised to -23.0 LUFS under EBU R 128 [38], jointly across clips, and the levels were verified by re-reading the rendered files. This is not cosmetic. The level error of the reconstruction reaches 9 dB against the original on some clips, and in an unnormalised test the loudest track wins the rating regardless of its quality. The reference administered to participants is therefore the raw dataset audio multiplied by a single scalar gain, with a residual at -78 dB, which is the requantisation noise of a sixteen-bit file: the same audio at a different level.

Blinding and randomisation. Participants saw only the labels A to E, in an order randomised independently for each clip, with no indication of which system produced which track.

The hidden reference. Including the unprocessed audio among the rated conditions is the only instrument available for detecting a participant who was not listening. It is also what makes post-screening possible at all, and the screening criterion was fixed before the responses were examined: a participant whose mean rating of the hidden reference falls below 4.0 on the five-point scale is excluded, on the grounds that failing to recognise uncompressed audio indicates either inattention or listening conditions in which no judgement is informative. Because a threshold chosen after seeing the outcome would be an elegant way of manufacturing one, the verdict is reported in Chapter 5 at five thresholds rather than at the declared one alone.

Participants and analysis

Fifty-six complete questionnaires were collected between the 25th and the 27th of August 2026, which is every response received before collection closed rather than a subset chosen afterwards: 1960 ratings, thirty-five per session, with no abandoned sessions and no missing cells. Participants are identified by session, the requested name being optional and non-identifying; no identifier of any kind appears in this thesis. The median age is 25 with a range of 20 to 61, the median session lasts 388 s, thirty-one listened on headphones and twenty-one on computer or tablet speakers, and thirty-seven described themselves as amateur musicians against fifteen non-experts and no experts. This is a convenience sample, young, largely non-expert and half of it listening on computer speakers; it is not a laboratory panel and is not presented as one.

The analysis is **within-subject**. Every participant rates every condition, so the difference between two systems computed within a participant cancels that participant's personal severity, which is the dominant source of variance: the mean rating given to the four degraded systems ranges from 1.61 to 4.0 across participants, an interval far wider than any difference between

systems. Confidence intervals are percentile bootstrap intervals over 10 000 resamples *of the participants*, who are the independent unit of observation, and not of the individual ratings, which are thirty-five correlated observations per person.

4.4 Probing the learned representation

The experiments described so far measure what the model reconstructs. This last group asks what its bottleneck actually contains, and it is exploratory in a sense the others are not: it was designed after the results of Section 4.1 were known, it produces no comparison between systems, and nothing in Chapters 5 to 7 depends on its outcome. It is included because a representation that is claimed to be efficient should be examined, not merely measured.

What is frozen, and what is trained

The model is never trained, modified or fine-tuned. The best checkpoint of the Ward arm is loaded, placed in evaluation mode, and every pass runs under no-gradient. The token embeddings are intercepted with a forward hook on the readout projection, which returns the mean of the posterior rather than a sample at evaluation time, so the representation is deterministic.

The only things trained anywhere in this section are **linear probes** — ridge regression and logistic regression — fitted *on top of* the frozen representation. A linear probe does not ask whether information could in principle be recovered; it asks whether it is present and linearly legible, which is the question worth asking of a code.

The analysis covers 525 clips of the test split, stratified by family with a ceiling of fifteen clips per instrument, giving 53 instruments across all ten families and roughly 227 000 tokens. Labels come from the official annotation: family, instrument, pitch, velocity, source, and ten hand-annotated timbral attributes.

Three design decisions that determine what can be concluded

The probe split is disjoint by instrument. Train and test partitions of the probe share no instrument. Without this, a probe memorises the instrument and every number is inflated; with it, a positive result means the representation generalises across instruments it has never seen.

Every probe is compared against a trivial control. The control is the same probe fitted on four coarse per-clip statistics — token count, mean log-magnitude, mean frame, mean fre-

quency bin — which any system would provide for free. A representation that does not beat those four numbers has not demonstrated that it contains anything.

Confounds are removed by residualisation rather than argued away. To ask whether a property is legible *independently* of pitch, the component of the representation linearly predictable from pitch is subtracted, with the regression estimated on the training clips alone. The same procedure is applied to level, in two versions: a single per-clip scalar, and the structured profile of mean log-magnitude in each frequency band by temporal half. The second is the severe one, and Section 5.2.1 reports why its result cannot be read the way it first appears.

The aggregation of tokens into a per-clip vector is itself a design variable and had to be treated as one. Averaging the roughly 430 tokens of a clip returns its mean level and destroys the spectral shape, which is where timbre lives; a first round of probes ran on that average and returned a negative result that belonged to the probe rather than to the model. The representation used throughout is therefore `bands+time`: active tokens averaged separately within six frequency bands and two temporal halves. Because richer representations also carry more free parameters, every comparison against the control is made at matched dimensionality, after a principal-component reduction estimated on the training clips only.

Two clustering methods, asking different questions

The geometry of the token cloud is examined with k-means and with HDBSCAN [39], and the pairing is deliberate. k-means partitions by construction: it assigns every point, assumes convex groups, and takes k from outside, so its output is the best available bipartition and never evidence that two groups exist. HDBSCAN builds a hierarchy across density scales, assumes no shape, and may declare a point to be noise, which is the vocabulary a continuum requires. A density-based method is necessary here rather than merely preferable: the median distance to the twentieth neighbour is 0.584 among silent tokens against 1.900 among active ones, so a single global radius cannot describe both regions.

Trajectories, and what the architecture forbids

The last group of experiments modifies token embeddings and decodes the result, which requires stating a limitation before any of its outcomes. **This model cannot generate.** Decoding needs five objects and the token embeddings are only one of them: the coarse graph, the map from the 128 763 grid nodes to tokens, the full-resolution graph with its weights, and the spectrogram shape all come from a real clip. This is the side information of Section 4.2 seen from

the generative side, and it is the same gap. What is possible is to borrow the structure of a real clip and move through the latent space above it: every sound produced is *this clip, if its content were that*, with the partition held fixed.

Finally, one property of the pipeline bounds every number in this section. The greedy coarsener turns last-bit differences in a merge cost into different discrete decisions, so the token count of a single clip is reproducible to no better than about five per cent across repeated passes on the same hardware. The averages reported elsewhere in this thesis are taken over hundreds of clips and are insensitive to it; an analysis that compares two passes over the *same* clip is not, and the trajectory experiments were run with deterministic reductions enabled for that reason.

5

Results

This chapter reports the results the thesis rests on, and it is divided in two. Section 5.1 concerns the model: what it reconstructs on the full corpus, how it compares against an external codec on a metric that is the objective of neither, and what fifty-two listeners said. It closes with what those results do not show. Section 5.2 then turns inward and asks what the bottleneck of the winning model came to contain; it is exploratory, and nothing established before it depends on what it finds.

The secondary measurements are not repeated here. The ablations that isolate each component, and the accounting of what it costs to transmit the partition, are reported in Chapter 4 beside the designs that produced them, because their function is to explain and to bound the results below rather than to constitute them.

Unless stated otherwise, the trained arms share encoder, decoder, objective and the weight $\beta = 10^{-3}$ of the variational bottleneck, and differ in one respect only: how the nodes of the time–frequency graph are assigned to tokens. Runs terminate by early stopping on the total validation loss and every figure is taken at the best checkpoint. Section 5.1.1 reports validation figures; everything from Section 5.1.2 onwards is measured on the held-out test split.

5.1 Results on the model

5.1.1 Reconstruction on the full corpus

The assignment of nodes to tokens

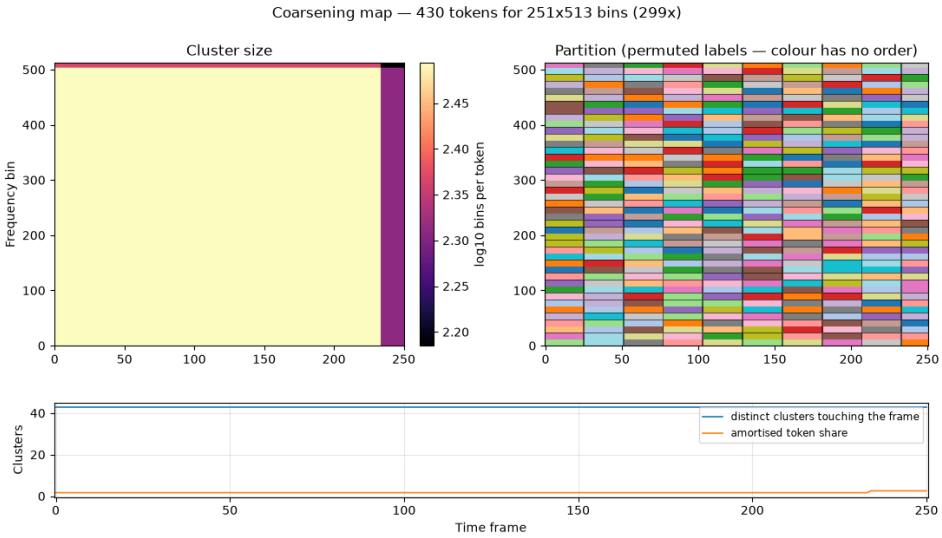
Table 5.1 compares the three assignment strategies at a comparable token count. The spread between the three arms is 2.5%, small enough that the comparison is about which partition is chosen and not about how much budget each is given.

Table 5.1: Node-to-token assignment on the full NSynth validation set, at a comparable token count. The three models share encoder, decoder, objective and $\beta = 10^{-3}$; only the way graph nodes become tokens differs. Adaptive coarsening under the total-variation criterion loses to fixed-grid pooling. Adding the cluster-size term to the merge cost reverses the verdict.

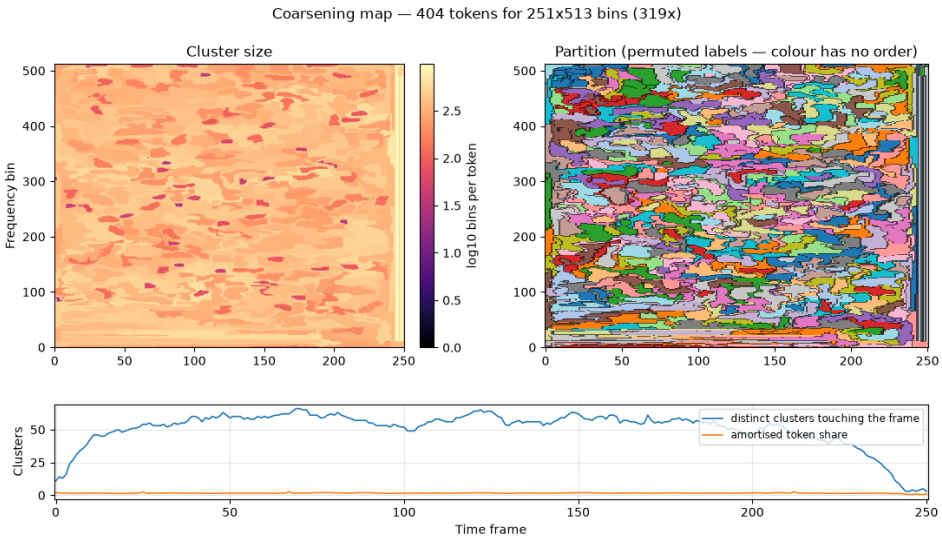
Assignment	Tokens	Compr.	Mel ↓	L_1 act. ↓	L_1 sil. ↓	KL (nats)
Uniform pooling (26×12)	430.0	300×	0.9560	0.7184	0.4985	14.43
Adaptive, total variation	422.2	305×	1.0274	0.7576	0.5627	**11.16**
Adaptive, Ward	**419.4**	**307×**	**0.6913**	**0.5149**	**0.3535**	11.98

The ordering found on the balanced subset survives the move to the full corpus, and it survives one further move: re-evaluating each best checkpoint on the held-out test split, where no selection took place, leaves the ranking unchanged and widens the matched-rate gap from 1.38 to 1.40 times. The selection bias described in Section 3.5 is therefore present and small, and it works against the result rather than for it. This is where the ordering matters most. Read as three rows the table gives one number, a mel distance 1.38 times better than the fixed grid. Read as a sequence it gives the result of this thesis, which is stronger and less comfortable: adaptivity by itself does not pay. Under the total-variation criterion the adaptive coarsener is 7% *worse* than a grid that ignores the content of the signal entirely, and only the size term of Section 3.2 reverses that. When the merge cost does not approximate the pooling error, the freedom to place cluster boundaries anywhere is spent badly and a rigid rule does better.

Two secondary observations support the reading. First, the improvement appears on both error terms, on active nodes and on silent ones alike, so it is not the product of reallocating budget from one region to the other. Second, the adaptive arm reaches this quality while transmitting 17% fewer nats through the bottleneck than the uniform arm: it is more accurate on less information, not more accurate on more.



(a) Uniform pooling, patch 26×12.



(b) Adaptive coarsening under Ward's linkage.

Figure 5.1: The partitions the two assignments actually produce, on one validation clip at a comparable token count. Left panels show how many time–frequency bins each token covers, right panels the partition itself; the labels of the right panels are permuted, so colour carries no order and only the boundaries are meaningful. The fixed grid produces cells of identical size everywhere and a constant number of clusters per frame, by construction. The adaptive coarsener produces cells that vary in size by more than an order of magnitude and whose shapes follow the structure of the signal, and its number of clusters per frame rises with the note and collapses over the silent tail. The distortion difference between these two partitions is the factor of 1.38 in Table 5.1.

The rate–distortion curve

The comparison at a matched token count answers a narrow question. The broader one is how much budget the fixed grid would need in order to reach the same quality, and Figure 5.2 answers it by measurement rather than by extrapolation.

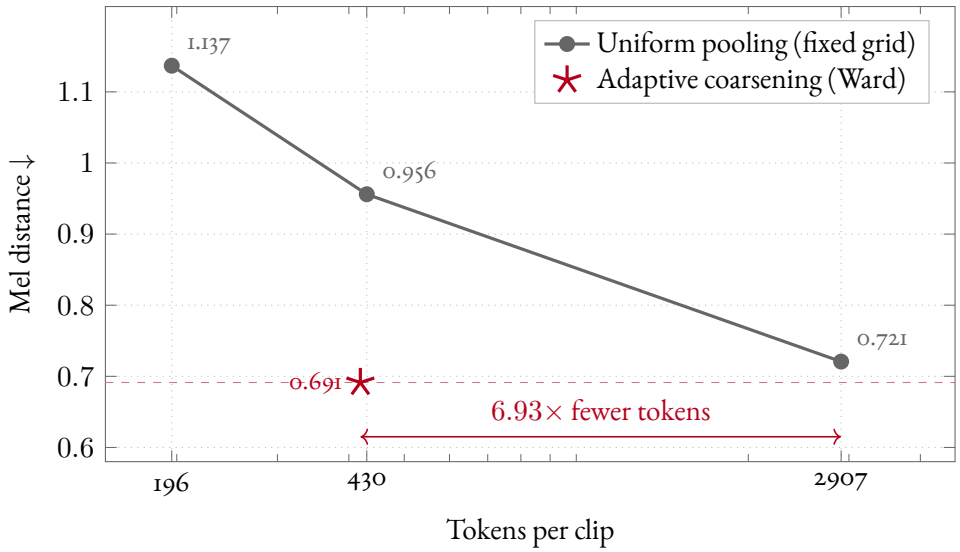


Figure 5.2: Rate–distortion curve of fixed-grid pooling, obtained by varying the time–frequency patch size, with the adaptive operating point superimposed. Both axes are as reported by validation on the full corpus; the horizontal axis is logarithmic. The adaptive method at 419 tokens per clip reaches a lower mel distance than uniform pooling at 2907 tokens, that is with 6.93 times fewer tokens. The factor is one in token count and not in transmitted bits; Section 4.2 accounts for the difference. The dashed line marks its distortion level, which the uniform curve does not reach anywhere in the measured range.

The adaptive point lies below the uniform curve at 6.93 times fewer tokens. The claim does not rest on fitting a curve and reading off an intersection: both configurations were trained and evaluated, and the fixed grid at 2907 tokens is still worse.

Two details are worth clarifying. The first concerns training budget: the adaptive run stopped after 23 epochs and the uniform run at 2907 tokens after 24, so the comparison is not an artefact of one model having been trained longer. The second concerns the subset used for the ablations: the same comparison performed there gave a factor of 7.0, and the fact that an experiment designed on a small balanced subset reproduces to within two per cent on the full corpus confirms the consistency of the result.

The same statement on the information axis

Table 5.2 restates the comparison on the second rate axis of Section 4.1, the information rate in nats supplied by the variational bottleneck.

Table 5.2: Distortion against information rate. The divergence of the variational bottleneck is a rate in nats that requires no quantisation. Uniform pooling spends $43\,401$ nats per clip without matching the quality that the adaptive method reaches with 5024.

Configuration	Tokens	KL/token	Nats/clip	Mel ↓
Uniform, patch 18×38	196	12.56	2 462	1.1368
Uniform, patch 26×12	430	14.43	6 205	0.9560
Uniform, patch 5×9	2907	14.93	43 401	0.7207
Adaptive, Ward	**419**	**11.98**	**5 024**	**0.6913**

The uniform arms differ from the adaptive one on both axes, and the ordering is preserved on both. That is what allows the word *efficient* to be used here without anything having been quantised, which was the purpose of introducing the bottleneck in the first place.

5.1.2 Distributional evaluation on the test split

The measurements in this section use the full test split of NSynth, 4096 clips whose instruments were seen neither in training nor in validation, and the Fréchet distance over CLAP embeddings described in Section 4.2. The metric is the training objective of none of the systems compared.

Table 5.3: Fréchet distance and paired cosine over CLAP embeddings, on the 4096 clips of the test split. The Griffin–Lim row receives the exact magnitude of the original and errs only in phase; it is the floor imposed by phase reconstruction, not a competing system.

System	Tokens/clip	FAD-CLAP ↓	Paired cosine ↑
Griffin–Lim (true magnitude)	—	0.011	0.979 ± 0.025
Adaptive, Ward	**433.1**	**0.156**	$\mathbf{0.807 \pm 0.092}$
Uniform 5×9	2907	0.179	0.789 ± 0.120
Uniform 26×12	430	0.289	0.688 ± 0.150
EnCodec @ $1.5\,\mathrm{kbit\,s^{-1}}$	—	0.301	0.667 ± 0.103
Uniform 18×38	196	0.318	0.646 ± 0.151
Adaptive, total variation	418.1	0.401	0.600 ± 0.160

One fact about the table is worth reading before the numbers in it. The two measurements, one distributional and one paired, agree on the entire ordering, which they need not have done: the first is blind to whether each reconstruction resembles *its own* original.

The internal comparison

Table 5.4: Ward against every other system. Differences in Fréchet distance, negative meaning Ward is closer to the originals, with 95% intervals over 200 bootstrap resamples evaluated on the same clips.

Against	Δ FAD	95% CI	Resamples won	Clips won
Adaptive, total variation	-0.2444	$[-0.2513, -0.2389]$	100%	96.5%
Uniform 18×38 (196 tok)	-0.1615	$[-0.1680, -0.1564]$	100%	89.0%
EnCodec @ $1.5\,\mathrm{kbit\,s^{-1}}$	-0.1451	$[-0.1498, -0.1408]$	100%	—
Uniform 26×12 (430 tok)	-0.1324	$[-0.1389, -0.1272]$	100%	82.1%
Uniform 5×9 (2907 tok)	-0.0226	$[-0.0268, -0.0182]$	100%	54.1%
Griffin–Lim (floor)	$+0.1455$	$[+0.1421, +0.1499]$	0%	—

Ward wins every internal comparison, with intervals that do not approach zero and every resample in its favour. The last row of the table is the one that bounds the achievement rather than celebrating it, and is discussed below.

Ward, with 433 tokens, is closer to the originals than the uniform grid with 2907: a factor of 6.7 in the number of tokens, on a metric that is the objective of neither. The comparison against the uniform arm at *matched* token count is far wider, at -0.1324 and 82.1% of clips. The same experiment on the balanced subset, with the training objective as the metric, gave a factor of 6.93; that two different metrics on two different splits agree to within four percent is strong evidence of the validity of adaptive coarsening.

The factor must be stated in the currency in which it was measured. **It is a factor in tokens, not in transmitted bits**, and Section 4.2 shows that in bits it very nearly vanishes. Nothing in this section supports the statement that this model compresses better than a neural codec.

The price of phase

The Griffin–Lim control receives the exact magnitude of the original and errs only in phase, and reaches a Fréchet distance of 0.011. Ward, at 0.156, sits fourteen times above that floor. This answers in advance the objection that the reconstructions are poor because of the phase reconstruction: they are not. The bottleneck is the coarsening.

The ratio is a ratio and not a decomposition. The Fréchet distance is not additive, so the error contributed by phase and the error contributed by coarsening are not independent quantities that sum. EnCodec is excluded from this comparison, since it reconstructs the waveform and this floor does not apply to it.

The external comparison, and what validates it

The validity check of Section 4.2 returns its verdict here.

Table 5.5: EnCodec at increasing rate, on the same clips, with the adaptive model placed on the same scale.

System	FAD-CLAP ↓	Paired cosine ↑
EnCodec @ $1.5\,\mathrm{kbit\,s^{-1}}$	0.3015	0.667 ± 0.103
EnCodec @ $3.0\,\mathrm{kbit\,s^{-1}}$	0.2319	0.735 ± 0.092
EnCodec @ $6.0\,\mathrm{kbit\,s^{-1}}$	0.1809	0.786 ± 0.083
Adaptive, Ward ($\sim$433 tokens)	**0.1562**	$\mathbf{0.807 \pm 0.092}$
EnCodec @ $12.0\,\mathrm{kbit\,s^{-1}}$	0.1463	0.820 ± 0.075
EnCodec @ $24.0\,\mathrm{kbit\,s^{-1}}$	0.1266	0.840 ± 0.070

The metric passes the test the log-mel distance had failed, which is to place this model at a plausible point on the rate scale of the codec. It falls monotonically with rate and places the adaptive model between 6 and $12\,\mathrm{kbit\,s^{-1}}$; log-linear interpolation puts the crossing at $9.8\,\mathrm{kbit\,s^{-1}}$ on the Fréchet distance and $9.2\,\mathrm{kbit\,s^{-1}}$ on the paired cosine, two independent measurements that agree. A metric that orders a codec correctly by its own rate can arbitrate this comparison; one that ranked this model ahead of the same codec at sixteen times the rate cannot.

The defensible statement is therefore that **on a perceptual metric of the magnitude domain, the adaptive model sits where EnCodec sits at around** $9\,\mathrm{kbit\,s^{-1}}$ **to** $10\,\mathrm{kbit\,s^{-1}}$ — and this is an *upper bound* rather than a point estimate. The audio encoder of CLAP takes 64 mel bands as input and never sees the waveform, hence never sees the phase. The proof is in the same table as the result: the Griffin–Lim condition, whose phase is destroyed outright, is judged nearly identical to the original. The gravest defect of this model is thus largely invisible to the metric, while EnCodec preserves phase and receives no credit for it. A metric that saw phase would move the crossing to a lower rate.

None of this affects the internal comparison, where all five arms discard phase identically.

The more informative reading of the external result is not that this model beats EnCodec. It is that **EnCodec at its lowest rate lands essentially where the uniform baseline lands** — 0.301

against 0.289 at matched token count. An independent system, built on entirely different principles, confirms the level at which the fixed-rate reference sits, and the advantage measured here is that of adaptive coarsening over uniform coarsening rather than of this architecture over that one.

5.1.3 The listening test

The measurements of Section 5.1.2 are statistical statements about an embedding space. This section reports what fifty-two listeners said, under the protocol of Section 4.3. Its most valuable outcome is not the ranking of the systems but the verdict it returns on the metric that produced that ranking.

The ranking, and the size of the margin

Table 5.6: Mean opinion score on the five-point scale, averaged over the seven clips within each participant and then over the fifty-two participants who passed the screening, with the difference from the adaptive model computed *within* each participant. Intervals are percentile bootstrap intervals over $10\,000$ resamples of the participants; the last column counts the participants preferring the adaptive model, ties excluded.

Condition	MOS	95% CI	Diff.	95% CI	Prefer
Original (hidden reference)	4.64	$[4.55, 4.71]$	-1.346	$[-1.500, -1.201]$	0%
Adaptive, Ward	**3.29**	$[3.16, 3.43]$	—		—
Uniform 26×12	3.11	$[2.98, 3.24]$	$+0.179$	$[+0.077, +0.280]$	69%
EnCodec @ $1.5\,\mathrm{kbit\,s^{-1}}$	3.00	$[2.84, 3.16]$	$+0.288$	$[+0.151, +0.426]$	65%
Adaptive, total variation	2.31	$[2.17, 2.45]$	$+0.981$	$[+0.885, +1.077]$	98%

The hidden reference at 4.64 out of five is what licenses the rest of the table: had the unprocessed audio landed among the degraded systems, no ordering below it would have meant anything. Four participants of fifty-six were excluded by the declared criterion, all four borderline rather than random, which is a low discard rate for an online test. Differences are computed within participants because the standard deviations of the raw scores are around half a point — larger than any gap between the degraded systems — and that variance is personal severity rather than disagreement.

All three intervals against a competing system exclude zero. Two readings matter more than the intervals. The first is that **EnCodec at its lowest rate lands on the uniform baseline**: 3.00 against 3.11, and the interval on that step, $[-0.008, +0.225]$, is the one comparison in the test

that is not resolved. The ladder therefore has four rungs and not five,

$$\text{original} \gg \text{Ward} > \{\,\text{uniform} \approx \text{EnCodec}\,\} \gg \text{total variation},$$

and the middle two must never be listed on separate lines without that qualification. An instrument sharing no assumptions with the Fréchet distance thus reproduces what Table 5.3 indicated: the advantage is that of adaptive coarsening over uniform coarsening, not of this architecture over a neural codec. The second reading is the size of the margin. **The advantage over the rate-matched uniform control is 0.179 points on a five-point scale**, with sixty-nine per cent of participants preferring the adaptive model and twenty-one per cent the uniform grid: a consistent preference and a small one, and the two adjectives are not separable. The mean also conceals heterogeneity across stimuli: the preference is decisive on two of the seven clips, absent on four and reversed on one, and it is concentrated on the stimuli where the uniform baseline fails most audibly. With seven clips, why it lands there is a question this test cannot answer.

The verdict was also recomputed at five screening thresholds rather than at the declared one alone. It is unchanged at every threshold up to and including the declared 4.0 — the difference is $+0.179$ whether 56 participants are retained or 52 — while the strictest cut, 4.5, discards twenty of the fifty-six and narrows it to $+0.107$ with an interval of $[-0.012,\ +0.222]$ that touches zero. The declared threshold was fixed before the responses were examined, so it is the one that counts; that the margin is small enough to depend on the choice is part of the result.

What the test says about the metric

Table 5.7: Correlation between the objective measurements and the mean opinion score over the thirty-five clip-by-condition cells. The right-hand block removes the seven cells of the hidden reference and is the one that matters.

	All 35 cells		Degraded systems only	
Measurement	Pearson	Spearman	Pearson	Spearman
Paired cosine, CLAP	$+0.867$	$+0.857$	$\mathbf{+0.787}$	$\mathbf{+0.748}$
Log-mel distance	-0.750	-0.688	-0.491	-0.408
Envelope distance	-0.695	-0.703	-0.509	-0.437

Seven clips in five conditions give thirty-five cells, each with a mean opinion score and an objective measurement, which is enough to ask whether the metric predicts the ear. The column to read is the right-hand one: with the reference included, a single obvious cell — highest

rating, perfect measurement — carries much of the fit, and a metric able only to recognise the original would appear excellent while being useless for the comparison this thesis needs.

With the reference removed, **the CLAP cosine holds at** $+0.787$ **while the log-mel distance collapses from** -0.750 **to** -0.491. The log-mel distance orders these cells largely because it recognises the original; precisely where it is needed, in separating systems that are all degraded, it loses half its predictive power. The decision taken in Section 4.2 on the strength of an argument about what each metric measures is thereby confirmed against human hearing, which is the strongest form that justification could take and the reason this test was run.

The confirmation has a boundary. Within a single clip, ranking the four degraded systems by cosine agrees with the human ranking at a mean Spearman correlation of $+0.686$ and never reaches $+1.00$ on any clip: much of the $+0.787$ comes from the spread between conditions rather than from precision within each stimulus. The metric is honest on sets and unreliable on individual cases, which is how it is used in Section 5.1.2 and how it should be used elsewhere.

Three instruments — fifty-two listeners on seven clips, and two statistical measurements on 4096 — therefore agree on the complete ordering of the four systems they share, through different mechanisms and on different samples. One divergence is worth recording rather than smoothing over: the Fréchet distance separates the uniform grid from EnCodec, 0.289 against 0.301, where the listening test cannot. They agree in direction, and only the statistical instrument has the resolution to call that step.

5.2 The latent space

5.2.1 What the bottleneck learned to contain

The preceding sections established that the adaptive partition reconstructs better at matched token count and that the partition itself is expensive to transmit. This one asks a different question, under the protocol of Section 4.4: what does the code contain? It is exploratory, it compares no systems, and none of the conclusions of this thesis rest on it. Its interest is that it explains, in the model's own terms, why the result of Section 5.1.1 happened.

A quarter of the capacity, and one dimension carrying half of it

The bottleneck of the trained model carries 12.35 nats per token, in agreement with the 11.98 logged during training, but it does not spread them evenly. **Eighty-seven of the** 128 **dimensions are dead**, in the sense that their posterior is indistinguishable from the prior below a

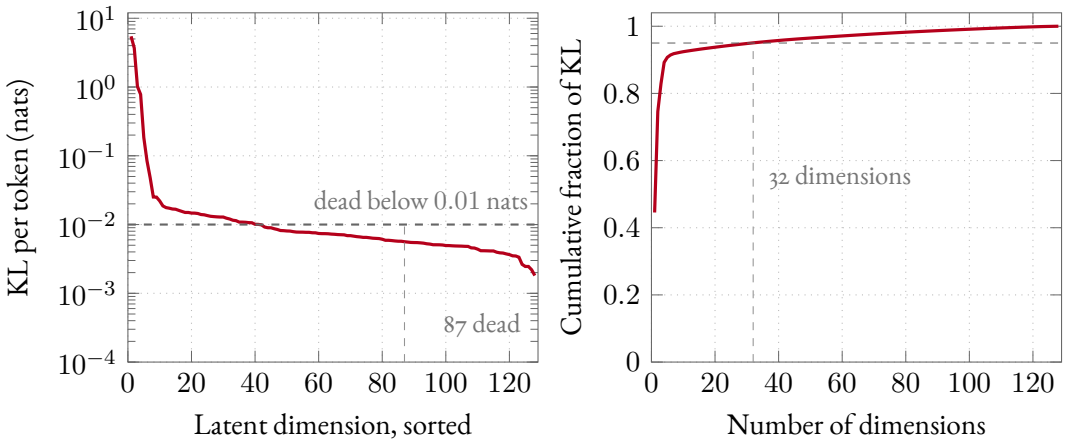


Figure 5.3: Kullback–Leibler divergence carried by each of the 128 latent dimensions, sorted in decreasing order on a logarithmic scale, with the cumulative fraction. Of the 128 dimensions, 87 have a posterior indistinguishable from the prior; 32 carry 95% of the information and a single one carries 45%. The dashed line on the left marks the threshold below which a dimension is counted as dead.

threshold of 0.01 nats; 32 dimensions suffice for 95% of the divergence, and **one dimension alone carries 45% of it**. Posterior collapse on this scale means the code uses a quarter of the capacity it was given, and every subsequent measurement in this section is computed on those 32 dimensions rather than on all 128. That choice is not cosmetic: taking instead the 98 dimensions that cover 99% of the divergence lets some sixty axes of pure prior noise dominate every Euclidean distance, which halves the signal in the measurements that follow.

What that dominant dimension encodes is legible directly. Its correlation with the mean log-magnitude of the token is 0.820 and with silence -0.673, against -0.347 with time and -0.154 with frequency: it is a level detector.

Asking the question dimension by dimension gives the same answer. Regressing each latent dimension on the observable factors, and averaging across dimensions *weighted by the divergence each carries*, log-magnitude explains 0.504 of the variance against 0.098 for the time–frequency position of the token and 0.004 for the size of the cluster it aggregates. Ranked by dominant factor, the dimensions governed by magnitude account for **88% of the information** in the code, and no dimension is governed by frequency. **This is a magnitude code**: it describes what is present rather than where it sits, which is the reading that supports calling the representation structure-aware, and it is the opposite of the outcome that was feared — a bottleneck spending its budget on describing the geometry of its own partition.

The word *content* should be resisted here, and so should *timbre*. The factor that wins is the log-magnitude, that is, the level.

The dominant axis is an audible gain

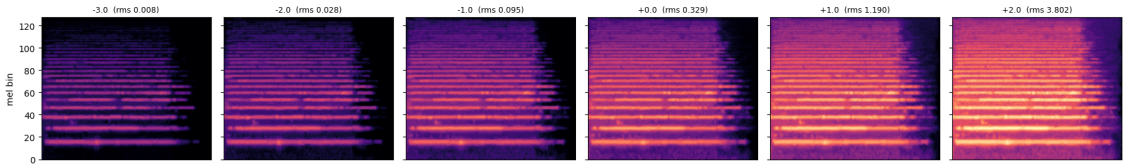


Figure 5.4: Mel spectrograms of the reconstruction as a constant δ is added to the most informative latent dimension of a single clip, with the partition and every other dimension held fixed. Each unit multiplies the amplitude by 3.45, constant over five steps: the dimension implements a logarithmic gain.

A correlation of 0.82 can be read; the same fact can also be heard. Adding a constant to the dominant dimension of one clip, and decoding with the partition and all other dimensions untouched, multiplies the amplitude of the result by 3.45 per unit — $10.8\,\mathrm{dB}$ — and the ratio holds across five consecutive steps. **Adding a constant in that space multiplies the amplitude**, so the model has learned a logarithmic gain on an axis.

Repeating the perturbation on the four most informative dimensions separates what they do. The first two carry 65% of the divergence between them and move the amplitude by factors of 142 and 24.9 over the same excursion; the third and fourth carry 20% and move it by 1.3 and 1.4, changing the sound while leaving the level almost fixed. **Level and content occupy separate axes, and level takes nearly the whole budget.**

The silence codeword

The founding intuition of this work was that silence should be cheap. The representation shows what the trained model did with it, without having been asked.

Labelling as silent every token whose mean log-magnitude falls below the threshold the coarsener itself uses, silent tokens are 56.3% of all tokens but account for only 30% of the latent variance: their mean squared distance from their own centroid is 0.39 times that of the active tokens, which corresponds to a root-mean-square radius 0.62 times as large. **A codeword for silence exists**, with two and a half times less variance than the rest of the code.

More telling than its width is its shape in the plane. The tokens of that group are nodes of the coarse graph, so their connectivity can be measured rather than assumed: within a single clip the silent group forms **on average** 1.1 **connected components**, against 43.4 for everything else. It is not dust scattered over the spectrogram; it is one region. And pulling all the tokens of a clip towards the centroid of its own silent tokens fades the sound monotonically to *exact* silence, with no gain touched anywhere.

This does not contradict the ablation of Section 4.1, which found that deliberately collapsing the silent subgraph never reaches the model that gives silence no special treatment. The two statements are complementary and together they sharpen the founding hypothesis rather than settling it either way: left alone, the model does concentrate silence into a compact, connected codeword; forced to do so by partitioning the graph in advance, it fragments the active region and loses. Silence is indeed cheap. It is the mechanism of reserving budget for it that fails.

A continuum, not a set of classes

Asking how many groups the code contains requires care about what each method can answer. On the 32 active dimensions, k-means is maximised at $k = 2$ with a silhouette of 0.339, no larger k being supported, and the smaller group is unambiguous: deep silence, mean log-magnitude -18.4, late in the clip, very large clusters. But k-means partitions by construction. That result is the best available **imposed bipartition** and is not evidence that two groups exist.

HDBSCAN, which measures density instead of imposing a partition, does not find them. At a minimum cluster size of ten per cent of the points it returns *no* stable cluster and labels everything noise; at a smaller setting it returns five clusters that are between 79 and 100% silent, while the half of the tokens it declares to be noise is the active region. A density-based method finds groups only where density is high, and here density is high only in silence. The defensible statement is therefore that **the code is a continuum without density modes in its active region, carrying two distinct populations of silent tokens**. Consistently with that, a ten-nearest-neighbour graph over $40\,000$ tokens places $38\,979$ of them in a single connected component.

The two populations of silence are worth separating, because the two methods found different ones. The group k-means isolates consists of the large merged blocks of the silent tail, averaging 613 nodes per token. The one HDBSCAN isolates — a filament holding 13.9% of all tokens, which k-means absorbs entirely into its majority cluster — consists of tokens of about *three* nodes: fragments that the coarsener failed to merge. These are an artefact of the partition made visible by the representation, and locating them suggests a concrete defect to repair rather than a property to report.

Timbre: a claim that fell, and one that held

Clips of *different instruments of the same family* land near one another in this space 2.89 times more often than chance, with neighbours from the same instrument excluded so that the measurement is about generalisation across instruments. It was the most striking number of the

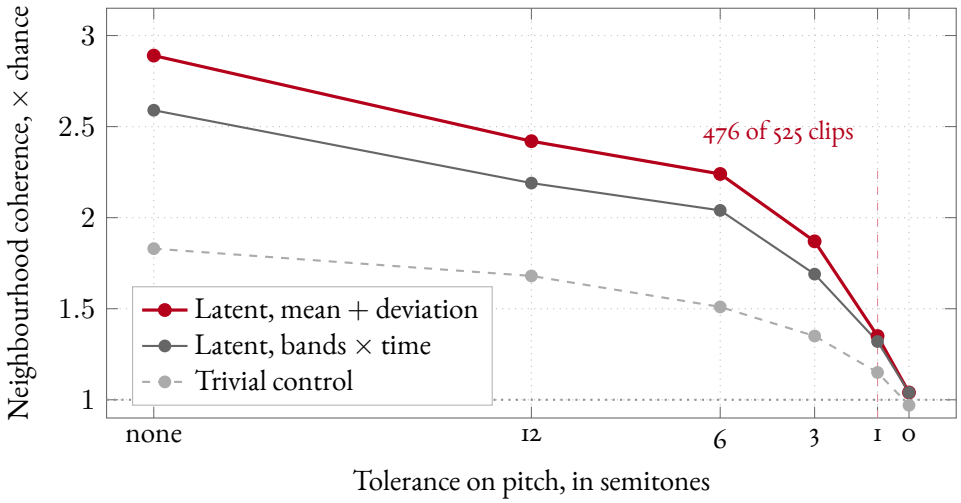


Figure 5.5: Timbral coherence of the neighbourhood as the tolerance on pitch is tightened: the fraction of a clip's ten nearest neighbours that share its family, divided by chance. Neighbours from the same instrument are excluded throughout, so the measurement is about generalisation across instruments, and the chance level is recomputed at each tolerance over the admissible candidates. The advantage of the representation over the trivial control falls from 2.89 times chance to 1.35: the coherence observed without the control was largely mediated by pitch. The rightmost point is not informative — at exact pitch only 67 clips remain and everything, control included, collapses to chance.

analysis, and it does not survive its control. Restricting the search to clips of similar pitch decays the advantage monotonically, and at one semitone of tolerance — 476 of the 525 clips, so not a small-sample effect — it reaches 1.35 against 1.15 for the trivial control. **Most of that signal was pitch.** A residue of timbral coherence at matched pitch remains, and it is modest.

One qualification belongs with this, and it is not an excuse. In this corpus family and pitch are entangled by construction: a bass is a bass partly because it is low. Matching pitch removes a confound and also removes a legitimate part of what family identity is. The control is still the right one — the real question is whether a flute is distinguished from a violin *on the same note* — but the collapse should be read knowing that part of it is a property of the label.

The claim that holds is a different one, and the distinction is the point. Hand-annotated timbral attributes, unlike family, are not determined by the note: a plucked sound decays quickly at any pitch. Probing those attributes on an instrument-disjoint split gives a mean area under the curve of 0.792, and **subtracting every direction linearly predictive of pitch costs** 0.004, leaving 0.788. That residualised representation beats a trivial control that contains the true pitch (0.736), which in turn beats pitch alone (0.609).

The mean conceals a mixed picture and should not be quoted alone. The representation wins outright on `nonlinear_env` and `bright`, where the control fails; it *loses* on `long_release`

Table 5.8: Timbral attributes, area under the curve on an instrument-disjoint split. The last two columns are the measurement that matters: removing pitch costs almost nothing, removing the structured level profile costs a great deal. The picture is mixed by attribute and the mean alone would misrepresent it. The first two columns come from the pitch control and the last three from the level control, run separately; the intact representation scores 0.792 in the first and 0.787 in the second, which is the run-to-run variation of the coarsener rather than a difference of protocol.

Attribute	Pitch only	Control + pitch	Latent	− pitch	− level
`bright`	0.554	0.484	0.764	0.755	0.693
`dark`	0.764	0.768	0.807	0.793	0.495
`distortion`	0.622	0.736	0.665	0.692	0.708
`fast_decay`	0.745	0.983	0.967	0.964	0.854
`long_release`	0.472	0.863	0.768	0.762	0.651
`multiphonic`	0.605	0.895	0.868	0.847	0.742
`nonlinear_env`	0.525	0.448	0.751	0.745	0.568
`percussive`	0.802	0.898	0.909	0.930	0.800
`reverb`	0.393	0.546	0.583	0.615	0.603
Mean	**0.609**	**0.736**	**0.787**	**0.789**	**0.679**

and `multiphonic`, where four coarse statistics plus the true pitch do better. It wins on average, and it wins most where the control has nothing. Getting this result at all required the representation to preserve joint time–frequency structure: with a plain average over tokens the effect disappears, and at matched dimensionality the band representation alone sits at the level of the control.

The confound that pitch was not

Pitch was the obvious confound to remove and it cost nothing. Level is the one that costs, and the reason is instructive rather than damaging.

Subtracting a single per-clip level scalar costs 0.027 of mean area under the curve. Subtracting the *structured* profile — mean log-magnitude in each of six bands by two temporal halves, that is, the same coordinates the representation is built on — costs 0.108, taking the mean from 0.787 to 0.679. Removing pitch and level together costs no more than level alone.

The lazy reading is that timbre was level in disguise and the claim falls. That reading is wrong, and the reason is physical rather than statistical. *Dark* means energy concentrated in the low bands; *bright* means the opposite; `fast_decay` means energy falling quickly. These attributes **are** distributions of level over time and frequency. For them the experiment does not ask the representation to read timbre without level; it asks it to predict a quantity after that quantity

has been removed, and the answer can only be no. `dark` duly falls to chance, 0.807 to 0.495.

The three attributes that are *not* definable as an energy profile — `distortion`, `reverb`, `multiphonic` — are the ones the control can be read on, and there the information survives intact, 0.705 to 0.684, with two of the three improving. What the data support is therefore this: **the representation reads timbral attributes through the time-frequency distribution of energy, which is largely what timbre consists of**; the information is independent of pitch and is not independent of level; and where level does not define the attribute, it survives.

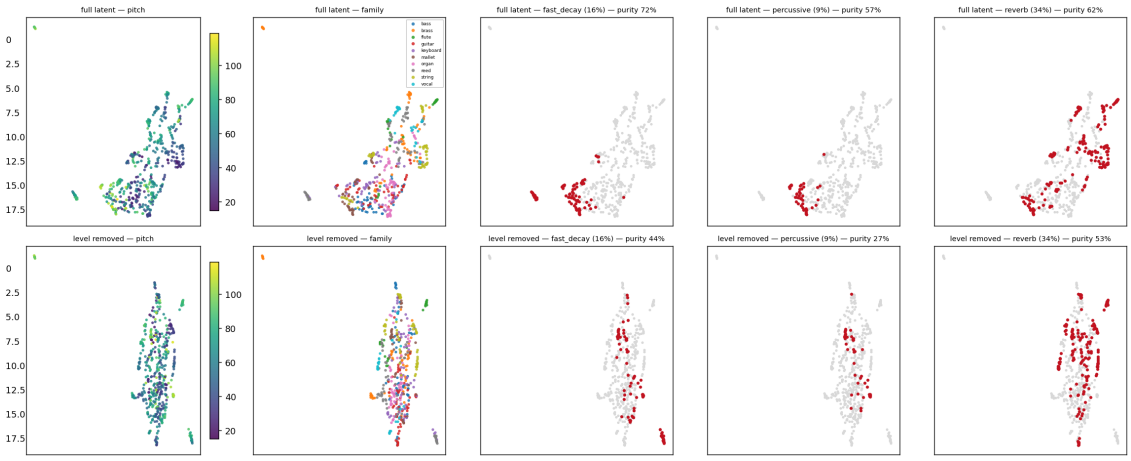


Figure 5.6: The same clips projected twice, with identical colourings: above, the representation as it is; below, the same representation with the profile of mean log-magnitude over frequency bands and temporal halves subtracted, which removes roughly three quarters of its variance. Pitch follows the level profile it is carried by: legible from the intact representation, and unreadable once that profile is gone, because pitch is largely *which bands hold energy*. Instrument family survives the removal. The three attribute panels show the two behaviours side by side: `fast_decay` and `percussive` form compact regions that disperse when level is taken away, while `reverb`, the one attribute that survives both controls, stays concentrated. The projection is stochastic and non-metric, so it illustrates the measurements in the text and does not stand in for them.

A second measurement, of geometry rather than of linear decodability, keeps this from being the whole story. Removing the level profile destroys roughly 74% of the variance of the representation, yet clips sharing an attribute still fall within one another's ten nearest neighbours at between 1.70 and 1.83 times the base rate, against 2.66 to 2.74 before removal. Ranges rather than single values are quoted because the coarsening is not deterministic: the figures are the spread over four repetitions of the whole measurement, and the separation between the two conditions is far wider than that spread. Level carried most of what was linearly legible, but it was not the only principle organising the space. The two measurements can even disagree on the same attribute: `distortion` is linearly separable and becomes *more* so once level is removed, while its neighbourhood purity sits below its own base rate — distorted clips are sep-

arable by a hyperplane and simultaneously not near one another. Only `reverb` survives both controls.

That same removal answers a question it was not asked. Pitch, which is legible from the intact representation at an R^2 between 0.34 and 0.41 across repetitions, becomes unreadable once the level profile is gone, while family accuracy is unchanged to within the variation between runs. The mechanism is the one that explains `dark`: pitch is largely *which bands hold energy*, so removing the band profile removes it by construction. This is a property of the removal and not a finding about the model, and the point is worth making explicitly, because the same figure read by eye suggested the opposite — the pitch gradient looks cleaner after removal than before. The projection is stochastic and non-metric, and the measurement overruled the picture.

The partition is specific to its signal

Interpolating from the tokens of one clip towards those of another, over the partition of the first, moves the reconstruction monotonically: the mel distance from the source rises from 0.955 to 9.824 and the distance from the target falls from 9.758 to 1.521. With $\beta = 10^{-3}$ the code is barely regularised towards a smooth prior, so this was not a given; the prediction made before running it was that it would fail.

The interesting number is where it stops. At the far end the distance from the target is 1.521 against the 0.327 of a genuine reconstruction of that target, a factor of nearly five, and the residual cannot be blamed on the correspondence between token sets: the method that performs best uses the exact node-level correspondence the shared grid provides and has no parameters at all. It is worth recording that the principled-looking alternative, a one-to-one optimal assignment between token sets, performs 53% *worse* — to give every token a distinct partner it must send some of them far away, and one-to-one is simply the wrong prior when two partitions have different granularity in different places.

What remains is the partition. The clusters of the source clip were built to fit the spectral structure *of that clip*; averaging another clip's content over them discards the structure that made it recognisable. **An adaptive partition is specific to its signal and does not transfer**, which is the central result of this thesis seen from the back: the coarsening wins because it fits the content, and for exactly that reason it does not fit a different one.

5.2.2 What this analysis is not

Every number in Section 5.2.1 comes from a single instrument-disjoint split of 525 clips, with no resampling. They are single measurements and carry no confidence intervals, so a difference of a few hundredths between two of them means nothing; the ones read as results here are the large ones, and the residual timbral coherence of 1.35 against 1.15 is explicitly not among them. The analysis was designed after the main results were known, which is legitimate for exploration and disqualifying for confirmation.

Three of its measurements also depend on choices that could reasonably have been made otherwise. The proportion of silent tokens follows the threshold the coarsener uses, and moves with it. The removal of the level profile is not a clean control for four of the nine attributes, because for them it removes part of the target. And the token count of an individual clip varies by about five per cent between identical passes, through the interaction of non-deterministic reductions with a greedy discrete merge — harmless for averages over hundreds of clips, which is how every other figure in this chapter is obtained, but a real constraint on any analysis of a single clip.

6 Discussion

The measurements are in Chapter 5. What they amount to is that a graph, coarsened by its own signal, is a workable substrate for representing music — workable in the sense that it reconstructs, that listeners prefer it to a content-blind control at matched rate, and that the mechanism responsible can be named. This chapter says what that establishes, what it opens, and where the account is still incomplete.

6.1 What the work establishes

Coarsening a graph is a viable way to build an audio representation, and this had not been shown. Graph neural networks had reached audio only for discriminative tasks, where phase is discarded because nothing must be reconstructed. Here a spectrogram of 128 763 nodes is contracted to roughly 420 tokens, expanded back, and resynthesised into sound that fifty-two listeners rate above a rate-matched fixed grid, above an external codec at its lowest rate, and above the earlier merge criterion — with three instruments sharing no assumptions ordering the systems identically. The construction works, and that is the first thing the thesis has to say.

At matched quality the representation is shorter, by a factor of 6.7. The adaptive partition reconstructs better than uniform pooling given seven times as many tokens, and the factor appears twice: under the training objective on the validation set and under a distributional metric on the held-out split that is the objective of neither system. Two measurements sharing no

assumptions agreeing to within four per cent is the strongest single piece of evidence in this work.

What produces the result is the merge criterion, and this is the finding most likely to travel. The same architecture spans both verdicts: under a cost that ignores how large the clusters being merged already are, the adaptive coarsener *loses* to the fixed grid it is meant to improve on; under a cost that weights the distance between cluster means by their sizes, and thereby approximates the error the subsequent pooling will commit, it wins. Adaptivity is not a property a method acquires by being described as adaptive. In this work it paid when the quantity minimised at each merge approximated the error that merge causes, and lost when it did not. Ward's cost does so exactly, since the costs of its merges sum to the within-cluster error of the final partition, which is why the result was to be expected rather than merely observed. The evidence is one criterion of each kind, so the rule is not established in general; whether it holds for other coarsening schemes, on other graphs, is the natural question it raises.

Two further quantities were measured rather than assumed. Removing message passing while holding depth, capacity, coarsener and decoder fixed raises the reconstruction error by 65%, so that the premise of the work now carries a number instead of an argument. And the metric on which the later comparisons rest was validated against human hearing: across the thirty-five cells of the listening test the paired cosine in a contrastive embedding predicts the human rating among degraded systems at $+0.787$, where the log-mel distance this project began with manages -0.491. That second result is the most transferable one here — not a fact about graphs, but a demonstration that the choice of metric in this domain is an empirical question with a checkable answer.

6.2 Why the graph was the right substrate

The justification for building on a graph is structural, and it is worth separating from the question of which edges were drawn, because the two are often conflated.

A grid can be subsampled, on a schedule chosen before any audio is seen. It cannot be *contracted unevenly*: there is no operation on a rectangular lattice that merges two neighbouring regions into one unit of the representation, keeps the object it produces of the same kind, and lets the same encoder and decoder run on the result. A graph has that operation natively. Everything this thesis measures — the merge criterion, the assignment of nodes to tokens, the comparison against a content-blind control — exists because that operation exists. The substrate is not decoration on a convolutional model; it is what makes the object of study expressible.

The analysis of the representation shows the same thing from the inside. Left to itself, the trained model gives silence a codeword of its own: silent tokens occupy a region with two and a half times less variance than active ones, and within a clip they form on average 1.1 connected components against 43.4 for everything else — one region, not dust scattered over the plane. Pulling the tokens of a clip towards that region fades the sound to exact silence with no gain touched anywhere. Nobody asked the model for this. It is what a coarsener free to contract where the signal is redundant converges to, and it is also why the token count comes down: the bottleneck spends itself on level and on silence, which is exactly what buys it the rate.

That reading also settles what the ablations mean. Sealing the silent region into a fixed number of super-nodes *before* coarsening never reaches the model that gives silence no special treatment, because a partitioned graph fragments the active region into several hundred components that can no longer be merged with one another — the partition degrades in shape before it degrades in size. Read beside the silence codeword the model builds on its own, the lesson is not about silence at all. **Structure of this kind is better induced than imposed**, and a graph is the substrate on which inducing it is possible.

The same distinction applies to the edges. Only the time–frequency lattice survives here; the harmonic prior reconstructs worse than the grid, reproducibly, because merging bins at different frequencies flattens the spectral envelope and misplaces the pitch. That is a result about one prior imposed by hand, not about the graph: propagation is worth 65% on the lattice alone, and the merging that carries the method operates on the lattice too. What the negative result marks is the boundary of what was tried, and the edge sets a graph would allow — learned rather than specified, induced from the signal rather than written down in advance — remain the most natural continuation of this work.

6.3 Two currencies, both honest

Two true statements about rate sit beside each other and answer different questions. At a matched token count the adaptive partition beats a fixed grid using 6.7 times as many tokens. In transmitted bits the advantage is between 1.0 and 1.6, because the decoder cannot recompute the node-to-token map and a fixed grid transmits its partition for nothing.

The question this thesis asks is the first. Its object is a representation, whose natural unit is the token: the length of the sequence is what a model downstream must process, and where such a representation is consumed rather than sent, the partition never crosses a channel at all — the encoder computes it and nothing is transmitted. The second figure is the price of the other

use, and no statement in this thesis claims that this model compresses better than a neural codec. What matters is that the cost was measured rather than assumed away, and that the estimate is an upper bound: the coarsening is a greedy hierarchy of merges, and a coder describing the sequence of merges rather than the resulting labelling could plausibly cost far less.

Getting the metric right was a precondition for all of it. The reconstruction metrics were sound within the family of models trained here and useless outside it — on the same clips the log-mel distance ranks this model ahead of an external codec at sixteen times its rate, while the signal-to-noise ratio ranks it behind by almost forty decibels — because one measures agreement in magnitude, which this model optimises, and the other agreement in waveform, which it discards. The replacement was admitted only after passing the test the previous one had failed, and then confirmed against fifty-two listeners. Building the instrument before trusting the reading is a large part of what this work consisted of.

6.4 The first stone

The ambition and the result are not the same size, and the gap is worth naming without treating it as a deficit.

The representation is adaptive in its *partition* and very nearly fixed in its *rate*: the coefficient of variation of the token count is between 0.09 and 0.12, measurably above the zero a fixed grid gives by construction and far below what the phrase variable-rate suggests. The configurations in which the rate does vary widely reconstruct badly, so within the range explored the two properties were not obtained together. The corpus is isolated notes rather than music in any ordinary sense, chosen because it presents the non-uniformity of information in an almost idealised form. Only one adaptive operating point was trained on the full corpus. Each of these bounds what can be claimed, and each is stated where the corresponding result appears.

None of them touches the mechanism. A time–frequency graph can be coarsened by a criterion the signal drives, inverted back into audio, and preferred by listeners to a content-blind control at the same rate; the criterion that decides the outcome has been identified and stated in transferable terms; the propagation has been given a number; and the code that results has been opened and described. The complete system has $654\,850$ parameters, twenty-two times fewer than the external codec it is measured against, and that figure is a design commitment rather than an apology.

What this leaves is a substrate with more room in it than has been used. The edges here are a lattice; a graph admits edge sets learned from the signal, and the one hand-specified prior

that was tried failed in a way that points at *how* rather than at *whether*. The rate here is nearly constant; sweeping the ceiling would turn a single operating point into a curve. The partition here is expensive to transmit; coding the merge hierarchy instead of its result is an unexplored and concrete route. None of these is a repair to a broken method. They are the next moves available on a board that did not exist before this work put a piece on it, and that is the sense in which the contribution should be read: a first stone, deliberately placed, on ground nobody had built on.

7 Conclusion

This thesis has shown that a graph neural network can serve as the encoder of an audio reconstruction model. The geometry of a graph is not that of a Euclidean space, and it is that freedom which allowed the representation to be contracted and expanded in a way a fixed grid cannot.

The main limitation of this work is that it uses only a small part of what the substrate permits: the edges here are a lattice, and the one hand-specified alternative did not survive its ablation. A graph, however, admits edge sets induced from the signal rather than written down in advance, and that space — which no other substrate offers — is where this work points next.

The strongest statement this thesis can make is that a model of roughly 650 000 parameters — twenty-two times smaller than the codec it was measured beside — earned a seat at what we like to call the adults' table.

References

[1] N. Zeghidour, A. Luebs, A. Omran, J. Skoglund, and M. Tagliasacchi, "SoundStream: An end-to-end neural audio codec," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 30, pp. 495–507, 2022.

[2] A. Défossez, J. Copet, G. Synnaeve, and Y. Adi, "High fidelity neural audio compression," *arXiv preprint arXiv:2210.13438*, 2022.

[3] R. Kumar, P. Seetharaman, A. Luebs, I. Kumar, and K. Kumar, "High-fidelity audio compression with improved RVQGAN," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[4] J. Engel, C. Resnick, A. Roberts, S. Dieleman, M. Norouzi, D. Eck, and K. Simonyan, "Neural audio synthesis of musical notes with WaveNet autoencoders," in *Proceedings of the International Conference on Machine Learning (ICML)*, 2017.

[5] A. Caillon and P. Esling, "RAVE: A variational autoencoder for fast and high-quality neural audio synthesis," *arXiv preprint arXiv:2111.05011*, 2021.

[6] M. Defferrard, X. Bresson, and P. Vandergheynst, "Convolutional neural networks on graphs with fast localized spectral filtering," in *Advances in Neural Information Processing Systems (NIPS)*, 2016.

[7] G. Karypis and V. Kumar, "A fast and high quality multilevel scheme for partitioning irregular graphs," *SIAM Journal on Scientific Computing*, vol. 20, no. 1, pp. 359–392, 1998.

[8] I. Higgins, L. Matthey, A. Pal, C. Burgess, X. Glorot, M. Botvinick, S. Mohamed, and A. Lerchner, "beta-VAE: Learning basic visual concepts with a constrained variational framework," in *International Conference on Learning Representations (ICLR)*, 2017.

[9] D. W. Griffin and J. S. Lim, "Signal estimation from modified short-time Fourier transform," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 32, no. 2, pp. 236–243, 1984.

[10] J. H. Ward, "Hierarchical grouping to optimize an objective function," *Journal of the American Statistical Association*, vol. 58, no. 301, pp. 236–244, 1963.

[11] Y. Ma, A. Øland, A. Ragni, B. M. Del Sette, C. Saitis, C. Donahue, C. Lin, C. Plachouras *et al.*, "Foundation models for music: A survey," *arXiv preprint arXiv:2408.14340*, 2024.

[12] X. Li, J. Liu, Y. Liang, Z. Niu, W. Chen, and X. Chen, "MeanAudio: Fast and faithful text-to-audio generation with mean flows," *arXiv preprint arXiv:2508.06098*, 2025.

[13] I. Rinaldi, M. Mendula, N. Fanelli, F. Levé, M. Testi, G. Castellano, and G. Vessio, "Art2Mus: Artwork-to-music generation via visual conditioning and large-scale cross-modal alignment," *arXiv preprint arXiv:2602.17599*, 2026.

[14] S. Li, Y. Li, Z. Wang, Y. Zhang, F. Wu, O. Deussen, T.-Y. Lee, and W. Dong, "SongEcho: Towards cover song generation via instance-adaptive element-wise linear modulation," in *International Conference on Learning Representations (ICLR)*, 2026.

[15] S. Wu, Z. Guo, R. Yuan, J. Jiang, S. Doh, G. Xia, J. Nam, X. Li, F. Yu, and M. Sun, "CLaMP 3: Universal music information retrieval across unaligned modalities and unseen languages," *arXiv preprint arXiv:2502.10362*, 2025.

[16] L. Ibáñez-Martínez, C. Nkama, A. Poltronieri, X. Serra, and M. Rocamora, "Evaluating disentangled representations for controllable music generation," *arXiv preprint arXiv:2602.10058*, 2026.

[17] S. Singh, C. J. Steinmetz, E. Benetos, H. Phan, and D. Stowell, "ATGNN: Audio tagging graph neural network," *IEEE Signal Processing Letters*, 2023, arXiv:2311.01526.

[18] A. Bhattacharjee, S. Singh, and E. Benetos, "GraFPrint: A GNN-based approach for audio identification," *arXiv preprint arXiv:2410.10994*, 2025.

[19] A. van den Oord, O. Vinyals, and K. Kavukcuoglu, "Neural discrete representation learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[20] M. Pasini, S. Lattner, and G. Fazekas, "CoDiCodec: Unifying continuous and discrete compressed representations of audio," in *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)*, 2025, arXiv:2509.09836.

[21] H. Zhang, Y. Guo, Z. Li, X. Hao, X. Chen, and K. Yu, "Unlocking temporal flexibility: Neural speech codec with variable frame rate," *arXiv preprint arXiv:2505.16845*, 2025.

[22] R.-C. Zheng, W. Liu, H.-P. Du, Q. Zhang, C. Deng, Q. Chen, W. Wang, Y. Ai, and Z.-H. Ling, "Say more with less: Variable-frame-rate speech tokenization via adaptive clustering and implicit duration coding," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026, arXiv:2509.04685.

[23] J. Li, Y. Qian, Y. Hu, L. Zhang, X. Wang, H. Lu, M. Thakker, J. Li, S. Zhao, and Z. Wu, "FlexiCodec: A dynamic neural audio codec for low frame rates," in *International Conference on Learning Representations (ICLR)*, 2026, arXiv:2510.00981.

[24] Q. Yang, L. He, B. Deng, and W. Zhao, "HARP: Harmonic-aware residual partitioning for neural audio codecs," *arXiv preprint arXiv:2607.16657*, 2026.

[25] Y. Cheng, X. Ma, G. Yu, and X. Gu, "Modeling music as a time–frequency image: A 2D tokenizer for music generation," *arXiv preprint arXiv:2605.15831*, 2026.

[26] D. I. Shuman, S. K. Narang, P. Frossard, A. Ortega, and P. Vandergheynst, "The emerging field of signal processing on graphs: Extending high-dimensional data analysis to networks and other irregular domains," *IEEE Signal Processing Magazine*, vol. 30, no. 3, pp. 83–98, 2013.

[27] A. Ortega, *Introduction to Graph Signal Processing*. Cambridge University Press, 2022.

[28] P. M. Djurić and C. Richard, Eds., *Cooperative and Graph Signal Processing: Principles and Applications*. Academic Press, Elsevier, 2018.

[29] A. Loukas, "Graph reduction with spectral and cut guarantees," *Journal of Machine Learning Research*, vol. 20, no. 116, pp. 1–42, 2019.

[30] A. Joly and N. Keriven, "Graph coarsening with message-passing guarantees," *arXiv preprint arXiv:2405.18127*, 2024.

[31] S. Elfwing, E. Uchibe, and K. Doya, "Sigmoid-weighted linear units for neural network function approximation in reinforcement learning," *arXiv preprint arXiv:1702.03118*, 2017.

[32] T. Cai, S. Luo, K. Xu, D. He, T.-Y. Liu, and L. Wang, "GraphNorm: A principled approach to accelerating graph neural network training," in *Proceedings of the International Conference on Machine Learning (ICML)*, 2021.

[33] X. Glorot and Y. Bengio, "Understanding the difficulty of training deep feedforward neural networks," in *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics (AISTATS)*, vol. 9, 2010, pp. 249–256.

[34] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.

[35] Y. Wu, K. Chen, T. Zhang, Y. Hui, T. Berg-Kirkpatrick, and S. Dubnov, "Large-scale contrastive language-audio pretraining with feature fusion and keyword-to-caption augmentation," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.

[36] A. Gui, H. Gamper, S. Braun, and D. Emmanouilidou, "Adapting Fréchet audio distance for generative music evaluation," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024, pp. 1331–1335.

[37] K. Kilgour, M. Zuluaga, D. Roblek, and M. Sharifi, "Fréchet audio distance: A reference-free metric for evaluating music enhancement algorithms," in *Interspeech*, 2019, pp. 2350–2354.

[38] European Broadcasting Union, "Loudness normalisation and permitted maximum level of audio signals," European Broadcasting Union, Geneva, EBU Recommendation R 128, 2023, version 5.0.

[39] R. J. G. B. Campello, D. Moulavi, and J. Sander, "Density-based clustering based on hierarchical density estimates," in *Advances in Knowledge Discovery and Data Mining (PAKDD)*, ser. Lecture Notes in Computer Science, vol. 7819. Springer, 2013, pp. 160–172.